

DATA COMMUNICATIONS AND COMPUTER NETWORKS NOTES

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OBJECTIVE:

Data communications and networking is the fastest growing technologies in our culture today. The course attempts

1. To provide a unified overview of the broad field of data and computer communications.
2. Emphasizes basic principles and topics of fundamental importance concerning the technology and architecture of this field
3. Provides a detailed discussion of leading edge topics.

CONTENTS

UNIT I: History and Overview

- 1.1. Introduction
- 1.2. Data Communication
- 1.3. Review Questions

UNIT II: Data Communications

- 2.1. Data Communications
- 2.2. Data Transmission
- 2.3. Transmission Impairments
- 2.4. Data Transmission Mode
- 2.5. Transmission Media

UNIT III: Communications Networks Architectures

- 3.1. Network Line Configuration
- 3.2. Network Devices
- 3.3. Network Topologies
- 3.4. Connection-Oriented and Connectionless Services

UNIT IV: Communications Networks Architectures

- 4.1. Network Protocol
- 4.2. TCP/IP Protocol Suite
- 4.3. Four Layers of TCP/IP Model
- 4.4. Open System Interconnection (OSI) Reference Model
- 4.5. Layers of the OSI Layers
- 4.6. Network Standards and Standardization Bodies
- 4.7. Review Questions

UNIT V: Local and wide area networks

- 5.1. LAN Technologies
- 5.2. Large Networks and Wide Area Networks
- 5.3. Types of WAN Technologies
- 5.4. Review Questions

UNIT VI: Client Server Computing

- 6.1. Web Technologies
- 6.2. Server-Side Programs
- 6.3. Socket Programming
- 6.4. Server Sockets
- 6.5. Review Questions

UNIT VII: Data Security and Integrity

- 7.1. Fundamentals of Network Security
- 7.2. Goals of Network Security
- 7.3. Cryptography
- 7.4. Types of Cryptosystems
- 7.5. Firewalls
- 7.6. Virtual Private Network
- 7.7. Review Questions

OUTCOMES

Upon completion of the subject, the student will be able to:

1. Understand unified overview of the broad field of data and computer communications.
2. Emphasizes basic principles and topics of fundamental importance concerning the technology Understand the architecture of this field
3. Learn detailed discussion of leading edge topics.

Textbook:

Computer Networking: A top down approach, 4th/5th edition, James F. Kurose and Keith W. Ross

References:

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UNIT - I

1.1. Introduction

Computer networks are the basis of communication in IT. They are used in a huge variety of ways and can include many different types of network. A computer network is a set of computers that are connected together so that they can share information. The earliest examples of computer networks are from the 1960s, but they have come a long way in the half-century since then.

A computer network comprises two or more computers that are connected either by cables (wired) or WiFi (wireless) with the purpose of transmitting, exchanging, or sharing data and resources. You build a computer network using hardware (e.g., routers, switches, access points, and cables) and software (e.g., operating systems or business applications).

Geographic location often defines a computer network. For example, a LAN (local area network) connects computers in a defined physical space, like an office building, whereas a WAN (wide area network) can connect computers across continents. The internet is the largest example of WAN, connecting billions of computers worldwide.

You can further define a computer network by the protocols it uses to communicate, the physical arrangement of its components, how it controls traffic, and its purpose.

Computer networks enable communication for every business, entertainment, and research purpose. The internet, online search, email, audio and video sharing, online commerce, livestreaming, and social networks all exist because of computer networks.

Computer network types

As networking needs evolved, so did the computer network types that serve those needs. Here are the most common and widely used computer network types:

- **LAN (local area network):** A LAN connects computers over a relatively short distance, allowing them to share data, files, and resources. For example, a LAN may connect all the computers in an office building, school, or hospital. Typically, LANs are privately owned and managed.
- **WLAN (wireless local area network):** A WLAN is just like a LAN but connection between devices on the network are made wirelessly.
- **WAN (wide area network):** As the name implies, a WAN connects computers over a wide area, such as from region to region or even continent to continent. The internet is the largest WAN, connecting billions of computers worldwide. You will typically see collective or distributed ownership models for WAN management.
- **MAN (metropolitan area network):** MANs are typically larger than LANs but smaller than WANs. Cities and government entities typically own and manage MANs.
- **PAN (personal area network):** A PAN serves one person. For example, if you have an iPhone and a Mac, it's very likely you've set up a PAN that shares and syncs content— text messages, emails, photos, and more—across both devices.

- **SAN (storage area network):** A SAN is a specialized network that provides access to block-level storage shared network or cloud storage that, to the user, looks and works like a storage drive that's physically attached to a computer. (For more information on how a SAN works with block storage, see Block Storage: A Complete Guide.)
- **CAN (campus area network):** A CAN is also known as a corporate area network. A CAN is larger than a LAN but smaller than a WAN. CANs serve sites such as colleges, universities, and business campuses.
- **VPN (virtual private network):** A VPN is a secure, point-to-point connection between two network end points (see 'Nodes' below). A VPN establishes an encrypted channel that keeps a user's identity and access credentials, as well as any data transferred, inaccessible to hackers.

The following are some common terms to know when discussing computer networking:

- **IP address:** An IP address is a unique number assigned to every device connected to a network that uses the Internet Protocol for communication. Each IP address identifies the device's host network and the location of the device on the host network. When one device sends data to another, the data includes a 'header' that includes the IP address of the sending device and the IP address of the destination device.
- **Notes:** A node is a connection point inside a network that can receive, send, create, or store data. Each node requires you to provide some form of identification to receive access, like an IP address. A few examples of nodes include computers, printers, modems, bridges, and switches. A node is essentially any network device that can recognize, process, and transmit information to any other network node.
- **Routers:** A router is a physical or virtual device that sends information contained in data packets between networks. Routers analyze data within the packets to determine the best way for the information to reach its ultimate destination. Routers forward data packets until they reach their destination node.
- **Switches:** A switch is a device that connects other devices and manages node-to-node communication within a network; ensuring data packets reach their ultimate destination. While a router sends information between networks, a switch sends information between nodes in a single network. When discussing computer networks, 'switching' refers to how data is transferred between devices in a network.

The three main types of switching are as follows:

- *Circuit switching*, which establishes a dedicated communication path between nodes in a network. This dedicated path assures the full bandwidth is available during the transmission, meaning no other traffic can travel along that path.
- *Packet switching* involves breaking down data into independent components called packets which, because of their small size, make fewer demands on the network. The packets travel through the network to their end destination.
- *Message switching* sends a message in its entirety from the source node, traveling from switch to switch until it reaches its destination node.
- **Ports:** A port identifies a specific connection between network devices. Each port is identified by a number. If you think of an IP address as comparable to the address of a hotel, then ports are the suites or room numbers within that hotel. Computers use port numbers to determine which application, service, or process should receive specific messages.

- **Network cable types:** The most common network cable types are Ethernet twisted pair, coaxial, and fiber optic. The choice of cable type depends on the size of the network, the arrangement of network elements, and the physical distance between devices.

Network Basic Understanding

A system of interconnected computers and computerized peripherals such as printers is called computer network. This interconnection among computers facilitates information sharing among them. Computers may connect to each other by either wired or wireless media.

1. **Network Engineering:** Networking engineering is a complicated task, which involves software, firmware, chip level engineering, hardware, and electric pulses. To ease network engineering, the whole networking concept is divided into multiple layers. Each layer is involved in some particular task and is independent of all other layers. But as a whole, almost all networking tasks depend on all of these layers. Layers share data between them and they depend on each other only to take input and send output.
2. **Internet:**

A network of networks is called an internetwork, or simply the internet. It is the largest network in existence on this planet. The internet hugely connects all WANs and it can have connection to LANs and Home networks. Internet uses TCP/IP protocol suite and uses IP as its addressing protocol. Present day, Internet is widely implemented using IPv4. Because of shortage of address spaces, it is gradually migrating from IPv4 to IPv6.

Internet enables its users to share and access enormous amount of information worldwide. It uses WWW, FTP, email services, audio and video streaming etc. At huge level, internet works on Client-Server model.

Internet uses very high speed backbone of fiber optics. To inter-connect various continents, fibers are laid under sea known to us as submarine communication cable.

Applications of Communication & Computer Network

Computer systems and peripherals are connected to form a network. They provide numerous advantages:

- Resource sharing such as printers and storage devices
- Exchange of information by means of e-Mails and FTP
- Information sharing by using Web or Internet
- Interaction with other users using dynamic web pages
- IP phones
- Video conferences
- Parallel computing
- Instant messaging

Characteristics of a Computer Network

- Share resources from one computer to another.
- Create files and store them in one computer, access those files from the other computer(s) connected over the network.
- Connect a printer, scanner, or a fax machine to one computer within the network and let other computers of the network use the machines available over the network.

Following is the list of hardware's required to set up a computer network.

- Network Cables
- Distributors
- Routers
- Internal Network Cards
- External Network Cards

A. Network Cables: Network cables are used to connect computers. The most commonly used cable is Category 5 cable RJ-45.



Figure 1. 1 category 5 network cable

B. Distributors

A computer can be connected to another one via a serial port but if we need to connect many computers to produce a network, this serial connection will not work.



Figure 1. 2 Central distributor

The solution is to use a central body to which other computers, printers, scanners, etc. can be connected and then this body will manage or distribute network traffic.

C. Router

A router is a type of device which acts as the central point among computers and other devices that are a part of the network. It is equipped with holes called ports. Computers and other devices are connected to a router using network cables. Now-a-days router comes in wireless modes using which computers can be connected without any physical cable.



Figure 1. 3 wifi router

D. Network Card

Network card is a necessary component of a computer without which a computer cannot be connected over a network. It is also known as the network adapter or Network Interface Card (NIC). Most branded computers have network card pre-installed. Network cards are of two types: Internal and External Network Cards.

E. Internal Network Cards

Motherboard has a slot for internal network card where it is to be inserted. Internal network cards are of two types in which the first type uses Peripheral Component Interconnect (PCI) connection, while the second type uses Industry Standard Architecture (ISA). Network cables are required to provide network access.



Figure 1. 4 : Network interface card

F. External Network Cards: External network cards are of two types: Wireless and USB based. Wireless network card needs to be inserted into the motherboard; however no network cable is required to connect to the network.

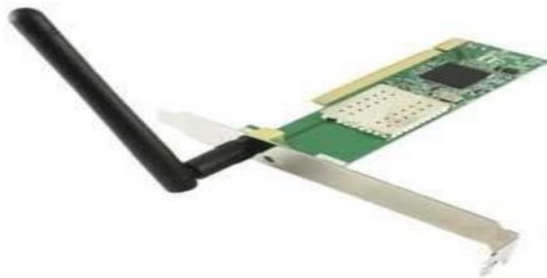


Figure 1. 5 External Network card

G. Universal Serial Bus (USB): USB card is easy to use and connects via USB port. Computers automatically detect USB card and can install the drivers required to support the USB network card automatically.



Figure 1. 6 Universal serial Bus

A system of interconnected computers and computerized peripherals such as printers is called computer network. This interconnection among computers facilitates information sharing among them. Computers may connect to each other by either wired or wireless media.

Classification of Computer Networks

Computer networks are classified based on various factors. They include:

- Geographical span
- Inter-connectivity
- Administration
- Architecture

A. Geographical Span

Geographically a network can be seen in one of the following categories:

- It may be spanned across your table, among Bluetooth enabled devices,. Ranging not more than few meters.
- It may be spanned across a whole building, including intermediate devices to connect all floors.
- It may be spanned across a whole city.
- It may be spanned across multiple cities or provinces.
- It may be one network covering whole world.

Personal Area Network

A Personal Area Network (PAN) is smallest network which is very personal to a user. This may include Bluetooth enabled devices or infra-red enabled devices. PAN has connectivity range up to 10 meters. PAN may include wireless computer keyboard and mouse, Bluetooth enabled headphones, wireless printers and TV remotes.

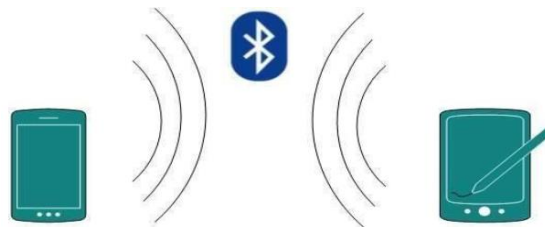


Figure 1. 7 Personal Area Network

For example, Piconet is Bluetooth-enabled Personal Area Network which may contain up to 8 devices connected together in a master-slave fashion.

Local Area Network

A computer network spanned inside a building and operated under single administrative system is generally termed as Local Area Network (LAN). Usually, LAN covers an organization' offices, schools, colleges or universities. Number of systems connected in LAN may vary from as least as two to as much as 16 million.

LAN provides a useful way of sharing the resources between end users. The resources such as printers; file servers scanners, and internet are easily sharable among computers.

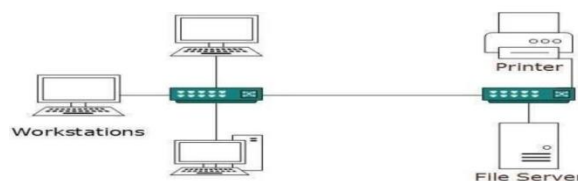


Figure 1. 8 Local Area Network

LANs are composed of inexpensive networking and routing equipment. It may contain local servers serving file storage and other locally shared applications. It mostly operates on private IP addresses and does not involve heavy routing. LAN works under its own local domain and controlled centrally.

LAN uses either Ethernet or Token-ring technology. Ethernet is most widely employed LAN technology and uses Star topology, while Token-ring is rarely seen. LAN can be wired, wireless, or in both forms at once.

Metropolitan Area Network

The Metropolitan Area Network (MAN) generally expands throughout a city such as cable TV network. It can be in the form of Ethernet, Token-ring, ATM, or Fiber Distributed DataInterface (FDDI).

Metro Ethernet is a service which is provided by ISPs. This service enables its users to expand their Local Area Networks. For example, MAN can help an organization to connect all of its offices in a city.

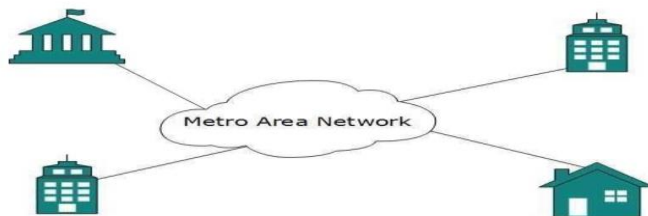


Figure 1. 9 Metropolitan Area Network

Backbone of MAN is high-capacity and high-speed fiber optics. MAN works in between Local Area Network and Wide Area Network. MAN provides uplink for LANs to WANs or internet.

Wide Area Network

As the name suggests, the Wide Area Network (WAN) covers a wide area which may span across provinces and even a whole country. Generally, telecommunication networks are Wide Area Network. These networks provide connectivity to MANs and LANs. Since they are equipped with very high speed backbone, WANs use very expensive network equipment.

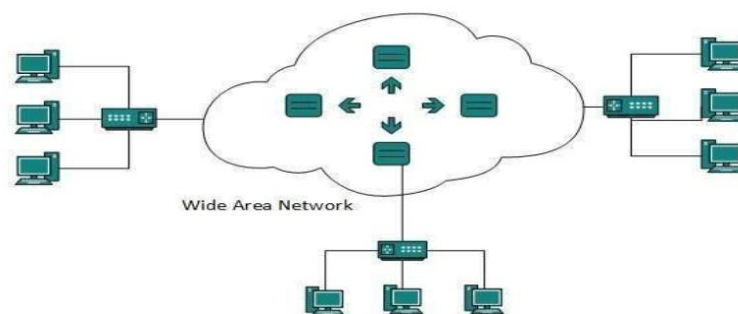


Figure 1. 10 Wide Area network

WAN may use advanced technologies such as Asynchronous Transfer Mode (ATM), Frame Relay, and Synchronous Optical Network (SONET). WAN may be managed by multiple administrations.

Internetwork

A network of networks is called an internetwork, or simply the internet. It is the largest network in existence on this planet. The internet hugely connects all WANs and it can have connection to LANs and Home networks. Internet uses TCP/IP protocol suite and uses IP as its addressing protocol. Present day, Internet is widely implemented using IPv4. Because of shortage of address spaces, it is gradually migrating from IPv4 to IPv6.

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Internet uses very high speed backbone of fiber optics. To inter-connect various continents, fibers are laid under sea known to us as submarine communication cable.

Internet is widely deployed on World Wide Web services using HTML linked pages and is accessible by client software known as Web Browsers. When a user requests a page using some web browser located on some Web Server anywhere in the world, the Web Server responds with the proper HTML page. The communication delay is very low.

Internet is serving many proposes and is involved in many aspects of life. Some of them are:

- Web sites
- E-mail
- Instant Messaging
- Blogging
- Social Media
- Marketing
- Networking
- Resource Sharing
- Audio and Video Streaming

A. Inter-Connectivity

Components of a network can be connected to each other differently in some fashion. By connectedness we mean either logically, physically, or both ways.

- Every single device can be connected to every other device on network, making the network mesh.
- All devices can be connected to a single medium but geographically disconnected, created bus like structure.
- Each device is connected to its left and right peers only, creating linear structure.
- All devices connected together with a single device, creating star like structure.
- All devices connected arbitrarily using all previous ways to connect each other, resulting in a hybrid structure.

B. Administration: From an administrator's point of view, a network can be private network which belongs a single autonomous system and cannot be accessed outside its physical or logical domain. A network can be public which is accessed by all.

C. Network Architecture: Computer networks can be discriminated into various types such as Client-Server, peer-to-peer or hybrid, depending upon its architecture.

- There can be one or more systems acting as Server. Other being Client, requests the Server to serve requests. Server takes and processes request on behalf of Clients.
- Two systems can be connected Point-to-Point, or in back-to-back fashion. They both reside at the same level and called peers.
- There can be hybrid network which involves network architecture of both the above types.

D. Network Applications: Computer systems and peripherals are connected to form a network. They provide numerous advantages:

- Resource sharing such as printers and storage devices
- Exchange of information by means of e-Mails and FTP
- Information sharing by using Web or Internet
- Interaction with other users using dynamic web pages
- IP phones

- Video conferences
- Parallel computing
- Instant messaging

Generally, networks are distinguished based on their geographical span. A network can be as small as distance between your mobile phone and its Bluetooth headphone and as large as the internet itself, covering the whole geographical world.

1.2. Data Communication

Data communications refers to the transmission of this digital data between two or more computers and a computer network or data network is a telecommunications network that allows computers to exchange data. The physical connection between networked computing devices is established using either cable media or wireless media. The best-known computer network is the Internet.

Data communications and networking are changing the way we do business and the way we live.

Business decisions have to be made ever more quickly, and the decision makers require immediate access to accurate information. Why wait a week for that report from Germany to arrive by mail when it could appear almost instantaneously through computer networks?

Businesses today rely on computer networks and internetworks. But before we ask how quickly we can get hooked up, we need to know how networks operate, what types of technologies are available, and which design best fills which set of needs.

The development of the personal computer brought about tremendous changes for Business, industry, science, and education. A similar revolution is occurring in data Communications and networking. Technological advances are making it possible for Communications links to carry more and faster signals. As a result, services are evolving to allow use of this expanded capacity. For example, established telephone services such as conference calling, call waiting, voice mail, and caller ID have been extended.

Research in data communications and networking has resulted in new technologies. One goal is to be able to exchange data such as text, audio, and video from all points in the world. We want to access the Internet to download and upload information quickly and accurately and at any time.

When we communicate, we are sharing information. This sharing can be local or remote. Between individuals, local communication usually occurs face to face, while remote communication takes place over distance. The term telecommunication, which includes telephony, telegraphy, and television, means communication at a distance (tele is Greek for “far”).

The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

1. Delivery: The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

2. Accuracy: The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

3. Timeliness: The system must deliver data in a timely manner. Data delivered late are useless.

In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

4. Jitter: Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or

video packets. For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

Data communications refers to the transmission of this digital data between two or more computers and a computer network or data network is a telecommunications network that allows computers to exchange data. The physical connection between networked computing devices is established using either cable media or wireless media. The best-known computer network is the Internet.

This tutorial should teach you basics of Data Communication and Computer Network (DCN) and will also take you through various advance concepts related to Data Communication and Computer Network.

1.3. Review Questions

- What is computer network?
- What is Data communication means?
- Discuss on History of Computer network?
- What is the Purpose of Computer network?

UNIT - II

2.1. Data Communications

Data Communications is the transfer of data or information between a source and a receiver. The source transmits the data and the receiver receives it. The actual generation of the information is not part of Data Communications nor is the resulting action of the information at the receiver.

Data Communication is interested in the transfer of data, the method of transfer and the preservation of the data during the transfer process.

The general Communication Model

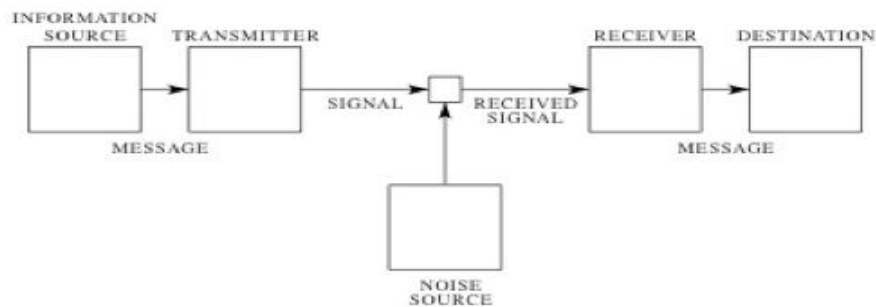


Figure 2. 1 Data communication model

1. An information source, which produces a message.
2. A transmitter, which encodes the message into signals
3. A channel, to which signals are adapted for transmission
4. A receiver, which 'decodes' (reconstructs) the message from the signal.
5. A destination, where the message arrives.

Activity 2.1

1. What is data transmission means and discuss in details
2. Discuss on analog and digital signals
3. How transmission impairment happen during data transmission?

2.2. Data Transmission: Data transmission is the process of sending digital or analog data over a communication medium to one or more computing, network, communication or electronic devices. It enables the transfer and communication of devices in a point-to-point, point-to-multipoint and multipoint-to multipoint environment.

Data transmission can be analog and digital but is mainly reserved for sending and receiving digital data. It works when a device or piece of equipment, such as a computer, intends to send a data object or file to one or multiple recipient devices, like a computer or server. The digital data originates from the source device in the form of discrete signals or digital bit streams. These data streams/signals are placed over a communication medium, such as physical copper wires, wireless carriers and optical fiber, for delivery to the destination/recipient device. Moreover, each outward signal can be baseband or passband.

In addition to external communication, data transmission also may be internally carried to a device. For example, the random access memory (RAM) or hard disk that sends data to a processor is also a form of data transmission.

Analog and Digital Transmission

Analog transmission is a method of conveying voice, data, image, signal, or video information. It uses a continuous signal varying in amplitude, phase, or another property that is in proportion to a specific characteristic of a variable. Analog transmission could mean that the transmission is a transfer of an analog source signal which uses an analog modulation method (or a variance of one or more properties of high frequency periodic waveform, also known as a carrier signal). FM and AM are examples of such a modulation. The transmission could also use no modulation at all. It is most notably an information signal that is constantly varying.

Data transmission (also known as digital transmission or digital communications) is a literal transfer of data over a point to point (or point to multipoint) transmission medium—such as copper wires, optical fibers, wireless communications media, or storage media. The data that is to be transferred is often represented as an electro-magnetic signal (such as a microwave). Digital transmission transfers messages discretely. These messages are represented by a sequence of pulses via a line code. However, these messages can also be represented by a limited set of wave forms that always vary. Either way, they are represented using a digital modulation method.

Analog transmission is capable of being conveyed in a no fewer than four ways: through a twisted pair or coax cable, through a fiber optic cable, through the air, or through water. There are, however, only two basic types of analog transmission. The first is known as amplitude modulation (or AM). This is a technique used in electronic communication and works by alternating the strength of a transmitted signal in relation to the information that is being sent. The second is known as frequency modulation (or FM). This type of communication conveys information over a carrier wave, just as AM transmission. However, FM communication alternates the frequency of the transmitted signal.

Data that is transmitted via digital transmission may be digital messages that have origins for a data source (a computer or a keyboard, for example). However, this transmitted data may also be from an analog signal (a phone call or a video signal, for example). It may then be digitized into a bit stream using pulse code modulation (or PCM)—or even more advanced source coding schemes. The coding of the data is carried out using codec equipment. There are a number of differences between analog and digital transmission, and it is important to understand how conversions between analog and digital occur. Let's look first at the older form of transmission, analog.

A. Analog Transmission

An analog wave form (or signal) is characterized by being continuously variable along amplitude and frequency. In the case of telephony, for instance, when you speak into a handset, there are changes in the air pressure around your mouth. Those changes in air pressure fall onto the handset, where they are amplified and then converted into current, or voltage fluctuations. Those fluctuations in current are an analog of the actual voice pattern—hence the use of the term *analog* to describe these signals.

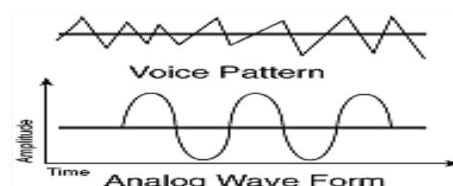


Figure 2. 2 Analog wave form

When it comes to an analog circuit what we also refer to as a voice-grade line—we need to also define the frequency band in which it operates. The human voice, for example, can typically generate frequencies from 100Hz

to 10,000Hz, for a bandwidth of 9,900Hz. But the ear does not require a vast range of frequencies to elicit meaning from ordinary speech; the vast majority of sounds we make that constitute intelligible speech fall between 250Hz and 3,400Hz. So, the phone company typically allotted a total bandwidth of 4,000Hz for voice transmission. Remember that the total frequency spectrum of twisted-pair is 1MHz. To provision a voice-grade analog circuit, bandwidth-limiting filters are put on that circuit to filter out all frequencies above 4,000Hz. That's why analog circuits can conduct only fairly low-speed data communications. The maximum data rate over an analog facility is 33.6Kbps when there are analog loops at either end.

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Analog facilities have limited bandwidth, which means they cannot support high-speed data. Another characteristic of analog is that noise is accumulated as the signal traverses the network.

As the signal moves across the distance, it loses power and becomes impaired by factors such as moisture in the cable, dirt on a contact, and critters chewing on the cable somewhere in the network. By the time the signal arrives at the amplifier, it is not only attenuated, it is also impaired and noisy. One of the problems with a basic amplifier is that it is a dumb device. All it knows how to do is to add power, so it takes a weak and impaired signal, adds power to it, and brings it back up to its original power level. But along with an increased signal, the amplifier passes along an increased noise level. So in an analog network, each time a signal goes through an amplifier, it accumulates noise. After you mix together coffee and cream, you can no longer separate them. The same concept applies in analog networks: After you mix the signal and the noise, you can no longer separate the two, and, as a result, you end up with very high error rates.

B. Digital Transmission

Digital transmission is quite different from analog transmission. For one thing, the signal is much simpler. Rather than being a continuously variable wave form, it is a series of discrete pulses, representing one bits and zero bits. Each computer uses a coding scheme that defines what combinations of ones and zeros constitute all the characters in a character set (that is, lowercase letters, uppercase letters, punctuation marks, digits, keyboard control functions).

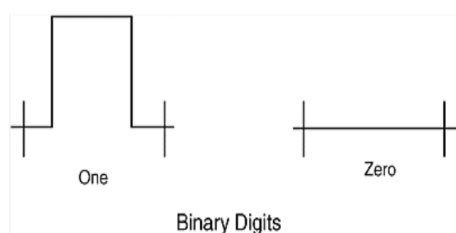


Figure 2.3 Digital transmission

How the ones and zeros are physically carried through the network depends on whether the network is electrical or optical. In electrical networks, one bits are represented as high voltage, and zero bits are represented as null, or low voltage. In optical networks, one bits are represented by the presence of light, and zero bits are represented by the absence of light. The ones and zeros—the on/off conditions—are carried through the network, and the receiving device repackages the ones and zeros to determine what character is being represented. Because a digital signal is easier to reproduce than an analog signal, we can treat it with a little less care in the network. Rather than use dumb amplifiers, digital networks use regenerative repeaters, also referred to as signal regenerators. As a strong, clean, digital pulse travels over a distance, it loses power, similar to an analog signal. The digital pulse, like an analog signal, is eroded by impairments in the network. But the weakened and impaired signal enters the regenerative repeater, where the repeater examines the signal to determine what was supposed to be a one and what was supposed to be a zero. The repeater regenerates a new

signal to pass on to the next point in the network, in essence eliminating noise and thus vastly improving the error rate.

2.3. Transmission Impairments

The signal received may differ from the signal transmitted. The effect will degrade the signal quality for analog signals and introduce bit errors for digital signals. There are three types of transmission impairments: attenuation, delay distortion, and noise.

(1) Attenuation: The impairment is caused by the strength of signals that degrades with distance over a transmission link. Three factors are related to the attenuation:

The received signal should have sufficient strength to be intelligently interpreted by a receiver. An amplifier or a repeater is needed to boost the strength of the signal.

A signal should be maintained at a level higher than the noise so that error will not be generated. Again, an amplifier or a repeater can be used.

Attenuation is an increasing function of frequency, with more attenuation at higher frequency than at lower frequency. An equalizer can smooth out the effect of attenuation across frequency bands, and an amplifier can amplify high frequencies more than low frequencies.

(2) Delay distortion: The velocity of propagation of a signal through a guided medium varies with frequencies; it is fast at the center of the frequency, but it falls off at the two edges of frequencies. Equalization techniques can be used to smooth out the delay distortion. Delay distortion is a major reason for the timing jitter problem, where the receiver clock deviates from the incoming signal in a random fashion so that an incoming signal might arrive earlier or late.

(3) Noise: Impairment occurs when an unwanted signal is inserted between transmission and reception. There are four types of noises:

- **Thermal noise:** This noise is a function of temperature and bandwidth. It cannot be eliminated. The thermal noise is proportional to the temperature and bandwidth as shown in the equation: thermal noise = $K(\text{constant}) * \text{temperature} * \text{bandwidth}$. Intermodulation noise this noise is caused by nonlinearity in the transmission system f_1 ; f_2 frequencies could produce a signal at $f_1 + f_2$ or $ABS(f_1 - f_2)$ and affect the frequencies at $f_1 + f_2$ or $ABS(f_1 - f_2)$.
- **Cross talk:** This type of noise is caused by electrical coupling in the nearby twisted pair or by unwanted signal picked by microwave antennas. For example, sometimes when you are on the telephone, you might hear someone else's conversation due to the cross talk problem.
- **Impulse noise:** Irregular pulses and short duration of relative high amplitude cause impulse noise. This noise is also caused by lightning and faults in the communication system. It is not an annoyance for analog data, but it is an annoyance for digital data. For example, 0.01 sec at 4800 bps causes 50 bits of distortion.

Activity 2.2

- Discuss and list some examples of half duplex transmission mode?
- Explain in details about parallel and serial data transmission mode?
- Discuss on the unguided transmission media using examples?
- Discuss some common guided transmission medium with examples?

2.4. Data Transmission Mode

Data

Transmission mode defines the direction of the flow of information between two communication devices. It is also called Data Communication or Directional Mode. It specifies the direction of the flow of information from one place to another in a computer network.

In the Open System Interconnection (OSI) Layer Model, the Physical Layer is dedicated to data transmission in the network. It mainly decides the direction of data in which the data needs to travel to reach the receiver system or node.

So, in this module, we will learn about different data transmission modes based on the direction of exchange, synchronization between the transmitter and receiver, and the number of bits sent simultaneously in a computer network.

According to the Direction of Exchange of Information:

1. Simplex

Simplex is the data transmission mode in which the data can flow only in one direction, i.e., the communication is unidirectional. In this mode, a sender can only send data but can not receive it. Similarly, a receiver can only receive data but cannot send it.

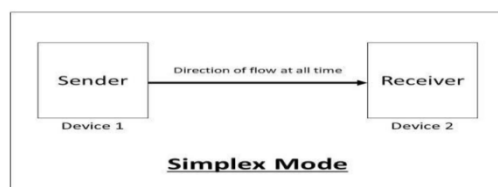


Figure 2. 4 Simplex transmission mode

This transmission mode is not so popular because we cannot perform two-way communication between the sender and receiver in this mode. It is mainly used in the business field as in sales that do not require any corresponding reply. It is similar to a one-way street. For Example, Radio and TV transmission, keyboard, mouse, etc.

2. Half-Duplex

Half-Duplex is the data transmission mode in which the data can flow in both directions but in one direction at a time. It is also referred to as Semi-Duplex. In other words, each station can both transmit and receive the data but not at the same time. When one device is sending the other can only receive and vice-versa.

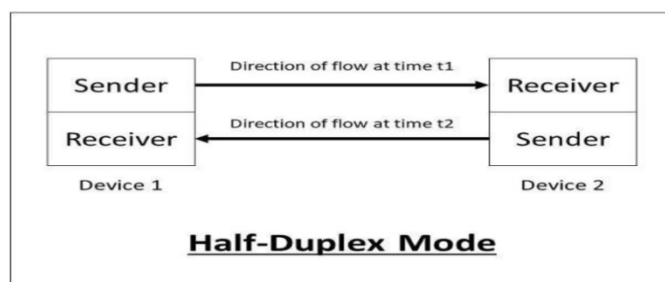


Figure 2. 5 half duplex transmission mode

In this type of transmission mode, the entire capacity of the channel can be utilized for each direction. Transmission lines can carry data in both directions, but the data can be sent only in one direction at a time.

This type of data transmission mode can be used in cases where there is no need for communication in both directions at the same time. It can be used for error detection when the sender does not send or the receiver does not receive the data properly. In such cases, the data needs to be transmitted again by the receiver. For Example, Police radio, Internet Browsers, etc.

3. Full-Duplex

Full-Duplex is the data transmission mode in which the data can flow in both directions at the same time. It is bi-directional in nature. It is two-way communication in which both the stations can transmit and receive the data simultaneously.

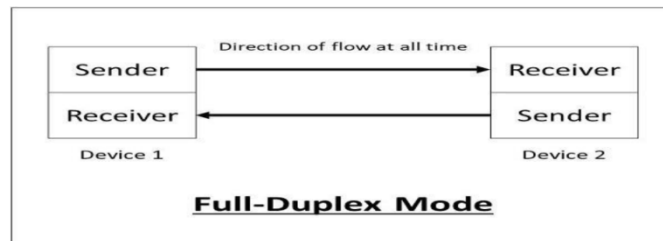


Figure 2. 6 Full duplex transmission mode

Full-Duplex mode has double bandwidth as compared to the half-duplex. The capacity of the channel is divided between the two directions of communication. This mode is used when communication in both directions is required simultaneously. For Example, a Telephone Network, in which both the persons can talk and listen to each other simultaneously.

According to the synchronization between the transmitter and the receiver:

1. Synchronous

The Synchronous transmission mode is a mode of communication in which the bits are sent one after another without any start/stop bits or gaps between them. Actually, both the sender and receiver are paced by the same system clock. In this way, synchronization is achieved.

In a Synchronous mode of data transmission, bytes are transmitted as blocks in a continuous stream of bits. Since there is no start and stop bits in the message block. It is the responsibility of the receiver to group the bits correctly. The receiver counts the bits as they arrive and groups them in eight bits unit. The receiver continuously receives the information at the same rate that the transmitter has sent it. It also listens to the messages even if no bits are transmitted.

In synchronous mode, the bits are sent successively with no separation between each character, so it becomes necessary to insert some synchronization elements with the message, this is called “Character-Level Synchronization”.

For Example, if there are two bytes of data, say (10001101, 11001011) then it will be transmitted in the synchronous mode as follows:

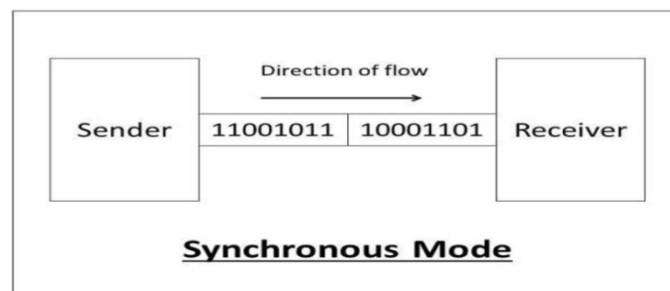


Figure 2. 7 Synchronous mode

For Example communication in CPU and RAM

2. Asynchronous

The Asynchronous transmission mode is a mode of communication in which a start and the stop bit is introduced in the message during transmission. The start and stop bits ensure that the data is transmitted correctly from the sender to the receiver.

Generally, the start bit is ‘0’ and the end bit is ‘1’. Asynchronous here means ‘asynchronous at the byte level’, but the bits are still synchronized. The time duration between each character is the same and synchronized.

In an asynchronous mode of communication, data bits can be sent at any point in time. The messages are sent at irregular intervals and only one data byte can be sent at a time. This type of transmission mode is best suited for short-distance data transfer.

For Example, if there are two bytes of data, say (10001101, 11001011) then it will be transmitted in the asynchronous mode as follows:

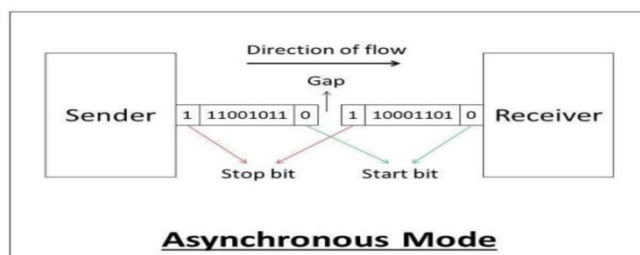


Figure 2. 8 Asynchronous mode

For Example, Data input from a keyboard to the computer.

According to the number of bits sent simultaneously in the network:

1. Serial

The Serial data transmission mode is a mode in which the data bits are sent serially one after the other at a time over the transmission channel.

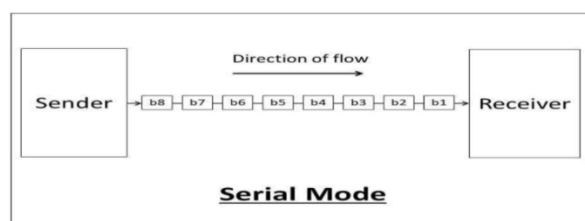


Figure 2. 9 Serial transmission mode

It needs a single transmission line for communication. The data bits are received in synchronization with one another. So, there is a challenge of synchronizing the transmitter and receiver.

In serial data transmission, the system takes several clock cycles to transmit the data stream. In this mode, the data integrity is maintained, as it transmits the data bits in a specific order, one after the other.

This type of transmission mode is best suited for long-distance data transfer, or the amount of data being sent is relatively small.

For Example, Data transmission between two computers using serial ports.

2. Parallel

The Parallel data transmission mode is a mode in which the data bits are sent parallelly at a time. In other words, there is a transmission of n-bits at the same time simultaneously.

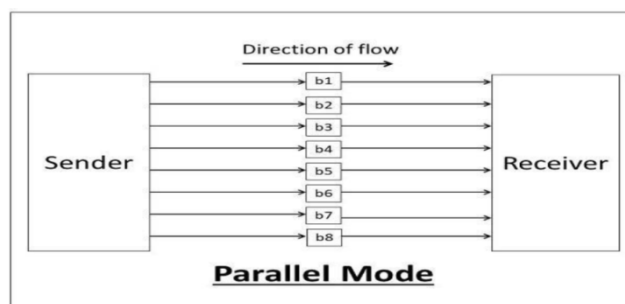


Figure 2. 10 Parallel transmission mode

Multiple transmission lines are used in such modes of transmission. So, multiple data bytes can be transmitted in a single system clock. This mode of transmission is used when a large amount of data has to be sent in a shorter duration of time. It is mostly used for short-distance communication.

For n-bits, we need n-transmission lines. So, the complexity of the network increases but the transmission speed is high. If two or more transmission lines are too close to each other, then there may be a chance of interference in the data, degrading the signal quality.

For Example, Data transmission between computer and printer.

Hence, after learning the various transmission modes, we can conclude that some points need to be considered when selecting a data transmission mode:

- Transmission Rate.
- The Distance that it covers.
- Cost and Ease of Installation.
- The resistance of environmental conditions.

This is all about the various transmission modes in a computer network.

2.5. Transmission Media

Communication channels that are used to carry the data from the transmitters to the receivers through the electromagnetic signals. The main function of this is to carry the data in the bits form through the Local Area Network (LAN). In data communication, it works like a physical path between the sender & the receiver. For instance, in a copper cable network the bits in the form of electrical signals whereas in a fiber network, the bits are available in the form of light pulses.

The quality, as well as characteristics of data transmission, can be determined from the characteristics of medium & signal. The properties of different transmission media are delay, bandwidth, maintenance, cost, and easy installation.

Transmission media is a pathway that carries the information from sender to receiver. We use different types of cables or waves to transmit data. Data is transmitted normally through electrical or electromagnetic signals. An electrical signal is in the form of current. An electromagnetic signal is series of electromagnetic energy pulses at various frequencies. These signals can be transmitted through copper wires, optical fibers, atmosphere, water and vacuum Different Medias have different properties like bandwidth, delay, cost and ease of installation and maintenance. Transmission media is also called Communication channel.

2.6. Types of Transmission Media: Transmission media is broadly classified into two groups.

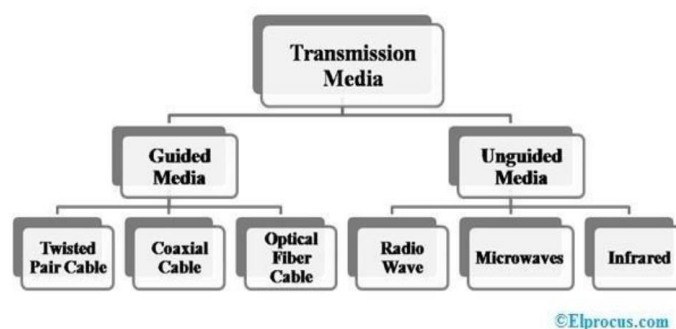


Figure 2: 11 Transmission media types

Guided Media

This kind of transmission media is also known as wired otherwise bounded media. In this type, the signals can be transmitted directly & restricted in a thin path through physical links. The main features of guided media mainly include secure, high-speed, and used in small distances. This kind of media is classified into three types which are discussed below.

A). Twisted Pair Cable

It includes two separately protected conductor wires. Normally, some pairs of cables are packaged jointly in a protective cover. Insulated copper wires arranged in regular spiral pattern.

- The oldest, least expensive, and most commonly used media
- reduce susceptibility to interference than straight pair wires (two straight parallel wires tend to act as an antenna and pick up extraneous signals when compared to twisted pairs)
- Highly susceptible to electrical noise, interference, and 'tapping' of the signal as compared to the other guided media
- Usually used for multiplexing multiple telephone lines, also used for transmitting digital data for point-to-point links (e.g. the leased line for AAUNet)
- Arrangement of twisted pairs into group used for high-speed (10-100 Mbps) LAN.



Figure 2. 12 Twisted pair cable

This is the most frequently used type of transmission media and it is available in two types.

1. UTP (Unshielded Twisted Pair)

This UTP cable has the capacity to block interference. It doesn't depend on a physical guard and used in telephonic applications. The advantage of UTP is a low cost, very simple to install, and high speed. The disadvantages of UTP is liable to exterior interference, transmits in fewer distances, and less capacity.

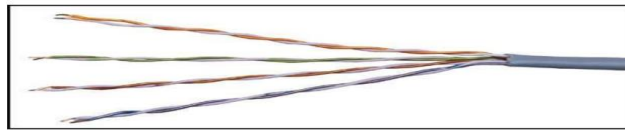


Figure 2. 13 Unshielded Twisted pair cable

Types of UTP

Category 3 Cable

With 10 MHz bandwidth, used for telco voice and horizontal wiring for 10-Mbps 10Base-T Ethernet or 4-Mbps Token Ring.

Category 4 Cable

With 20 MHz bandwidth, used for 16-Mbps Token Ring.

Category 5 Cable

The single most popular flavor! With 100 MHz bandwidth, it can handle upto 100-Mbps.

2. Shielded Twisted Pair

STP cable includes a particular jacket for blocking outside interference. It is used in rapid data rate Ethernet, in voice & data channels of telephone lines.

The main advantages of STP cable mainly include good speed, removes crosstalk.

The main disadvantages are hard to manufacture as well as install, It is expensive and bulky also

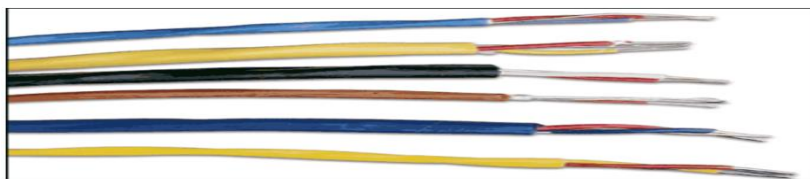


Figure 2. 14 Shielded Twisted pair cable

B) Coaxial Cable

This cable contains an external plastic cover and it includes two parallel conductors where each conductor includes a separate protection cover. This cable is used to transmit data in two modes like baseband mode as well as broadband mode. This cable is widely used in cable TVs & analog TV networks.

The advantages of the coaxial cable include high bandwidth, noise immunity is good, low cost and simple to install. The disadvantage of this cable is, the failure of cable can disturb the whole network.

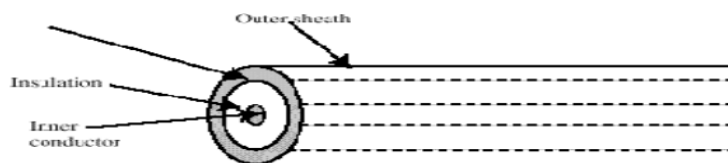


Figure 2. 15 Coaxial cable

- Most versatile medium used in LANs, Cable TV, VCR-to-TV connections
- Noise immunity is better than twisted pair
- Less susceptible to interference and cross talk but there still is attenuation and thermal noise problem
- Can go up to 185m (10Base2) or 500m(10Base5) without the need for an amplifier/repeater

C) Optical Fiber Cable

This cable uses the notion of light reflected through a core that is made with plastic or glass. The core is enclosed with less thick plastic or glass and it is known as the cladding, used for large volume data transmission.

The main advantages of this cable include lightweight, capacity & bandwidth will be increased, signal attenuation is less, etc. The disadvantages are high cost, fragile, installation & maintenance is difficult and unidirectional.

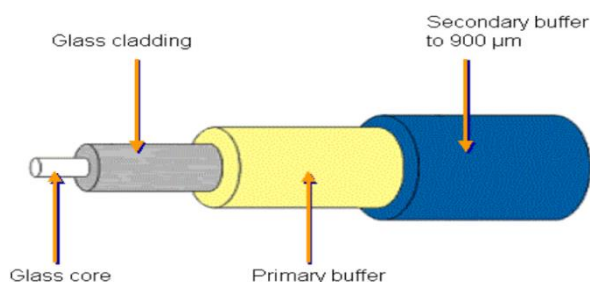


Figure 2. 16 : Optical fiber cable

Two types of fiber optic cables

Multimode Fiber optic cable

- Fiber optic cable where the light signal travels dispersed through the core
- Core is usually 50-62µm in diameter
- Maximum distance signal travels without a repeater is 500m

"Multimode fiber"
multiple paths through the fiber

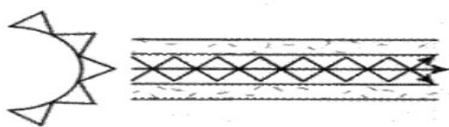


Figure 2. 17 Multimode fiber optic cable

Single Mode fiber

- Fiber optic cable where the light signal travels in a single mode through the core
- m in diameter μ Core is usually less than 10
- Maximum distance signal travels without a repeater is 10km (with the appropriate modulation up to 100km)

"Single mode fiber"
single path through the fiber

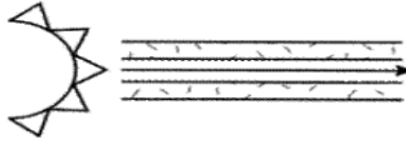


Figure 2. 18 Single mode fiber optic cable

Unguided (wireless transmission)

In unguided media transmission and reception are achieved by means of an antenna. It is also known as unbounded otherwise wireless transmission media. It doesn't require any physical medium to transmit electromagnetic signals. The main features of this media are less secure; the signal can be transmitted through air, and applicable for large distances. There are three types of unguided media which are discussed below.

A). Radio waves

These waves are very easy to produce as well as penetrate through buildings. In this, the transmitting & receiving antennas no need to align. The frequency range of these waves ranges from 3 kHz to 1GHz. These waves are used in AM & Fm radios for transmission. These waves are classified into two types namely Terrestrial & Satellite.

1. Terrestrial Microwave

- Typically used where laying a cable is not practical
- Parabolic dish shaped antenna for directional and bar-like antenna for omnidirectional transmission
- transmits/receives electromagnetic waves in the 2-40 GHz range
- Travels in a straight line (line-of-sight propagation)
- High data rates: 100's Mbps
- Repeaters spaced 10 – 100 km apart
- Applications : telephone and data transmission- wireless LANs

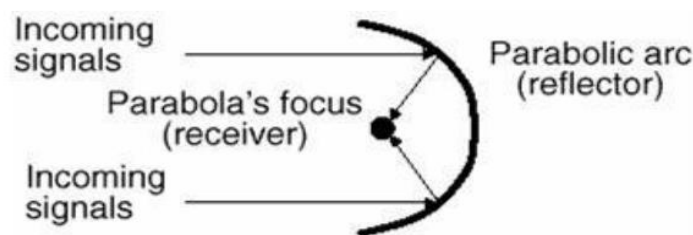


Figure 2. 19 Terrestrial microwave

2. Satellite Microwave

- Uses satellite in geostationary (geosynchronous) 36,000 km) ≈orbit(
- Source transmits signal to satellite which amplifies or repeats it, and retransmits down to destinations
- Optimum transmission in 1 - 10 GHz range;
- Bandwidth of 100's MHz
- 270ms Significant propagation delay about
- VSAT (Very small Aperture Terminal) :- High speed data transmission using satellite

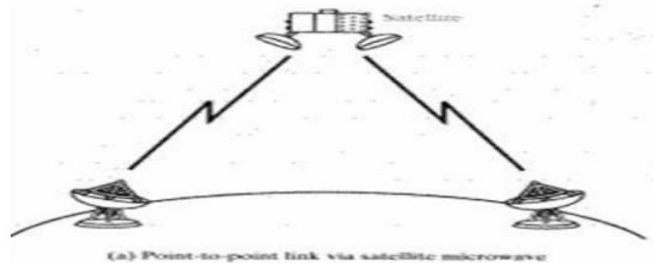


Figure 2. 20 Satellite microwave

B) Microwaves

It is a sightline transmission which means the transmitting & receiving antennas need to align correctly with each other. The distance which is covered through the signal can be directly proportional to the antenna's height. The frequency range of microwaves ranges from 1GHz to 300GHz. These are extensively used in TV distribution & mobile phone communication.

C) Infrared Waves

Infrared (IR) waves are used in extremely small distance communication as they cannot go through obstacles. So it stops intrusion between systems. The range of frequency of these waves is 300GHz to 400THz. These waves are used in TV remotes, keyboards, wireless mouse, printer, etc.

For short-range communication

- Remote controls for TVs, VCRs, and stereos
- Indoor wireless LANs

2.7. Review Questions

- How Data Communication works?
- List and explain Guided transmission mediums?
- What is transmission impairments means?
- How analog signals transmitted?
- In which means signals transmit without guided medium?
- How satellites provide data to the ground station?

UNIT - III

3.1. Network Line Configuration

Line configuration refers to the way two or more communication devices attached to a link. Line configuration is also referred to as connection. A Link is the physical communication pathway that transfers data from one device to another. For communication to occur, two devices must be connected in same way to the same link at the same time.

There are two possible line configurations.

1. Point-to-Point.
2. Multipoint.

Point-to-Point

A Point to Point Line Configuration Provide dedicated link between two devices use actual length of wire or cable to connect the two end including microwave & satellite link. Infrared remote control & tvs remote control.

The entire capacity of the channel is reserved for transmission between those two devices. Most point-to-point line configurations use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.

Point to point network topology is considered to be one of the easiest and most conventional network topologies. It is also the simplest to establish and understand. To visualize, one can consider point to point network topology as two phones connected end to end for a two way communication.

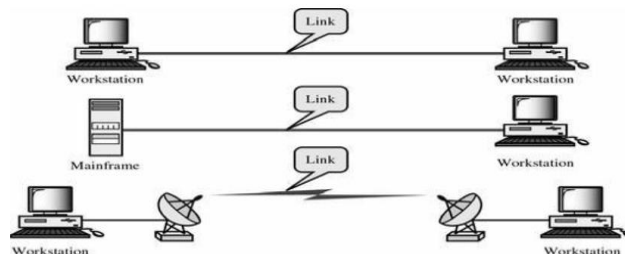


Figure 3. 1 Point to point network line Configuration

Multipoint Configuration

Multipoint Configuration also known as Multidrop line configuration one or more than two specific devices share a single link capacity of the channel is shared.

More than two devices share the Link that is the capacity of the channel is shared now. With shared capacity, there can be two possibilities in a Multipoint Line Config:

- Spatial Sharing: If several devices can share the link simultaneously, its called Spatially shared line configuration
- Temporal (Time) Sharing: If users must take turns using the link , then it's called Temporally shared or Time Shared Line Configuration

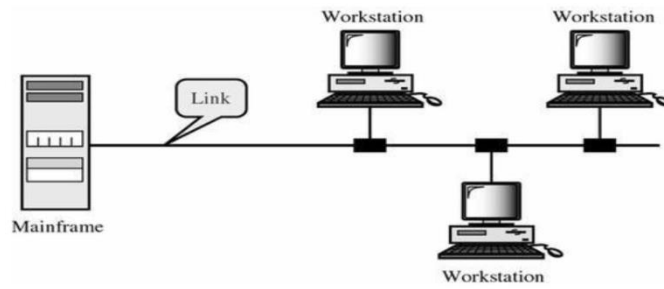


Figure 3. 2: Multiple point network line configurations

Activity 3.1

- Discuss on the network devices?
- What are the central devices among the network devices?
- How to differentiate Switch and Hub network devices?
- Discuss in details about functions of router?

3.2. Network Devices

Network devices, or networking hardware, are physical devices that are required for communication and interaction between hardware on a computer network

1. Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2 port device.

2. Hub – Hubs connect multiple computer networking devices together. A hub also acts as a repeater in that it amplifies signals that deteriorate after traveling long distances over connecting cables. A hub is the simplest in the family of network connecting devices because it connects LAN components with identical protocols.

A hub can be used with both digital and analog data, provided its settings have been configured to prepare for the formatting of the incoming data. For example, if the incoming data is in digital format, the hub must pass it on as packets; however, if the incoming data is analog, then the hub passes it on in signal form. Hubs do not perform packet filtering or addressing functions; they just send data packets to all connected devices. Hubs operate at the Physical layer of the Open Systems Interconnection (OSI) model. There are two types of hubs: simple and multiple port.

A hub is basically a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and Wastage.

- **Active Hub:** – These are the hubs that have their own power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.
- **Passive Hub:-** These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.
- **Intelligent Hub:-** It works like active hubs and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

3. Bridge – A bridge operates at the data link layer. A bridge is a repeater with add on the functionality of filtering content by reading the MAC addresses of source and destination.

It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

Types of Bridges

- **Transparent Bridges:-** These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.
- **Source Routing Bridges:-** In these bridges, routing operation is performed by the source station and the frame specifies which route to follow. The host can discover the frame by sending a special frame called the discovery frame, which spreads through the entire network using all possible paths to the destination.

4. Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but broadcast domain remains the same. Switches generally have a more intelligent role than hubs. A switch is a multiport device that improves network efficiency.

The switch maintains limited routing information about nodes in the internal network, and it allows connections to systems like hubs or routers. Strands of LANs are usually connected using switches. Generally, switches can read the hardware addresses of incoming packets to transmit them to the appropriate destination.

Using switches improves network efficiency over hubs or routers because of the virtual circuit capability. Switches also improve network security because the virtual circuits are more difficult to examine with network monitors.

You can think of a switch as a device that has some of the best capabilities of routers and hubs combined. A switch can work at either the Data Link layer

or the Network layer of the OSI model. A multilayer switch is one that can operate at both layers, which means that it can operate as both a switch and a router. A multilayer switch is a highperformance device that supports the same routing protocols as routers.

Switches can be subject to distributed denial of service (DDoS) attacks; flood guards are used to prevent malicious traffic from bringing the switch to a halt. Switch port security is important so be sure to secure switches: Disable all unused ports and use DHCP snooping, ARP inspection and MAC address filtering.

5. Routers –

Routers help transmit packets to their destinations by charting a path through the sea of interconnected networking devices using different network topologies. Routers are intelligent devices, and they store information about the networks they're connected to. Most routers can be

configured to operate as packet-filtering firewalls and use access control lists (ACLs). Routers, in conjunction with a channel service unit/data service unit (CSU/DSU), are also used to translate from LAN framing to WAN framing. This is needed because LANs and WANs use different network protocols. Such routers are known as border routers. They serve as the outside connection of a LAN to a WAN, and they operate at the border of your network. Router is also used to divide internal networks into two or more sub networks. Routers can also be connected internally to other routers, creating zones that operate independently. Routers establish communication by maintaining tables about destinations and local connections. A router contains information about the systems connected to it and where to send requests if the destination isn't known. Routers usually communicate routing and other information using one of three standard protocols: Routing Information Protocol (RIP), Border Gateway Protocol (BGP) or Open Shortest Path First (OSPF).

Routers are your first line of defense, and they must be configured to pass only traffic that is authorized by network administrators. The routes themselves can be configured as static or dynamic. If they are static, they can only be configured manually and stay that way until changed. If they are dynamic, they learn of other routers around them and use information about those routers to build their routing tables.

Routers are general-purpose devices that interconnect two or more heterogeneous networks. They are usually dedicated to special-purpose computers, with separate input and output network interfaces for each connected network. Because routers and gateways are the backbone of large computer networks like the internet, they have special features that give them the flexibility and the ability to cope with varying network addressing schemes and frame sizes through segmentation of big packets into smaller sizes that fit the new network components. Each router interface has its own Address Resolution Protocol (ARP) module, its own LAN address (network card

address) and its own Internet Protocol (IP) address. The router, with the help of a routing table, has knowledge of routes a packet could take from its source to its destination. The routing table, like in the bridge and switch, grows dynamically. Upon receipt of a packet, the router removes the packet headers and trailers and analyzes the IP header by determining the source and destination addresses and data type, and noting the arrival time. It also updates the router table with new addresses not already in the table. The IP header and arrival time information is entered in the routing table. Routers normally work at the Network layer of the OSI model.

A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets. Routers divide broadcast domains of hosts connected through it.

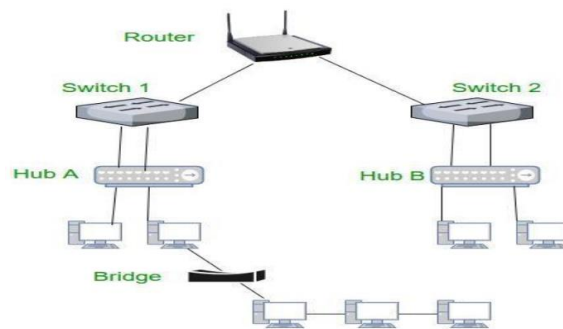


Figure 3. 3 Router network device

6. Gateway: Gateways normally work at the Transport and Session layers of the OSI model. At the Transport layer and above, there are numerous protocols and standards from different vendors; gateways are used to deal with them. Gateways provide translation between networking technologies such as Open System Interconnection (OSI) and Transmission Control Protocol/Internet Protocol (TCP/IP). Because of this, gateways connect two or more autonomous networks, each with its own routing algorithms, protocols, topology, domain name service, and network administration procedures and policies.

Gateways perform all of the functions of routers and more. In fact, a router with added translation functionality is a gateway. The function that does the translation between different network technologies is called a protocol converter.

7. Modem

Modems (modulators-demodulators) are used to transmit digital signals over analog telephone lines. Thus, digital signals are converted by the modem into analog signals of different frequencies and transmitted to a modem at the receiving location. The receiving modem performs the reverse transformation and provides a digital output to a device connected to a

modem, usually a computer. The digital data is usually transferred to or from the modem over a serial line through an industry standard interface, RS-232. Many telephone companies offer DSL services, and many cable operators use modems as end terminals for identification and recognition of home and personal users. Modems work on both the Physical and Data Link layers.

8. Access Point

While an access point (AP) can technically involve either a wired or wireless connection, it commonly means a wireless device. An AP works at the second OSI layer, the Data Link layer, and it can operate either as a bridge connecting a standard wired network to wireless devices or as a router passing data transmissions from one access point to another.

Wireless access points (WAPs) consist of a transmitter and receiver (transceiver) device used to create a wireless LAN (WLAN). Access points typically are separate network devices with a built-in antenna, transmitter and adapter. APs use the wireless infrastructure network mode to

provide a connection point between WLANs and a wired Ethernet LAN. They also have several ports, giving you a way to expand the network to support additional clients. Depending on the size of the network, one or more APs might be required to provide full coverage. Additional APs are used to allow access to more wireless clients and to

expand the range of the wireless network.

Each AP is limited by its transmission range — the distance a client can be from an AP and still obtain a usable signal and data process speed. The actual distance depends on the wireless standard, the obstructions and environmental conditions between the client and the AP. Higher end APs have high-powered antennas, enabling them to extend how far the wireless signal can travel. APs might also provide many ports that can be used to increase the network's size, firewall capabilities and Dynamic Host Configuration Protocol (DHCP) service. Therefore, we get APs that are a switch, DHCP server, router and firewall.

9. NIC

NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and router or modem. NIC card is a layer 2 device which means that it works on both physical and data link layer of the network model.

Activity 3.2

- Discuss on Connectionless and Connection oriented network?
- Discuss in details on logical and physical topologies?
- Compare and contrast the four main physical topologies

3.3. Network Topologies

Network topology refers to how various nodes, devices, and connections on your network are physically or logically arranged in relation to each other. Think of your network as a city, and the topology as the road map. Just as there are many ways to arrange and maintain a city—such as making sure the avenues and boulevards can facilitate passage between the parts of town getting the most traffic—there are several ways to arrange a network. Each has advantages and disadvantages and depending on the needs of your company, certain arrangements can give you a greater degree of connectivity and security.

There are two approaches to network topology: physical and logical. Physical network topology, as the name suggests, refers to the physical connections and interconnections between nodes and the network—the wires, cables, and so forth. Logical network topology is a little more abstract and strategic, referring to the conceptual understanding of how and why the network is arranged the way it is, and how data moves through it.

The way a network is arranged can make or break network functionality, connectivity, and protection from downtime. The question of, “What is network topology?” can be answered with an explanation of the two categories in the network topology.

1. Physical – The physical network topology refers to the actual connections (wires, cables, etc.) of how the network is arranged. Setup, maintenance, and provisioning tasks require insight into the physical network.

2. Logical – The logical network topology is a higher-level idea of how the network is set up, including which nodes connect to each other and in which ways, as well as how data is transmitted through the network. Logical network topology includes any virtual and cloud resources.

Effective network management and monitoring require a strong grasp of both the physical and logical topology of a network to ensure your network is efficient and healthy.

What is Star Topology?

A star topology, the most common network topology, is laid out so every node in the network is directly connected to one central hub via coaxial, twisted-pair, or fiber-optic cable. Acting as a server, this central node manages data transmission—as information sent from any node on the network has to pass through the central one to reach its destination—and functions as a repeater, which helps prevent data loss.

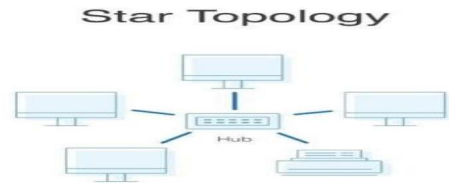


Figure 3. 4 Star network topology

Advantages of Star Topology

Star topologies are common since they allow you to conveniently manage your entire network from a single location. Because each of the nodes is independently connected to the central hub, should one go down, the rest of the network will continue functioning unaffected, making the star topology a stable and secure network layout? Additionally, devices can be added, removed, and modified without taking the entire network offline.

Disadvantages of Star Topology

On the flipside, if the central hub goes down, the rest of the network can't function. But if the central hub is properly managed and kept in good health, administrators shouldn't have too many issues. The overall bandwidth and performance of the network are also limited by the central node's configurations and technical specifications, making star topologies expensive to set up and operate.

What is Bus Topology?

A bus topology orients all the devices on a network along a single cable running in a single direction from one end of the network to the other—which is why it's sometimes called a "line topology" or "backbone topology." Data flow on the network also follows the route of the cable, moving in one direction.

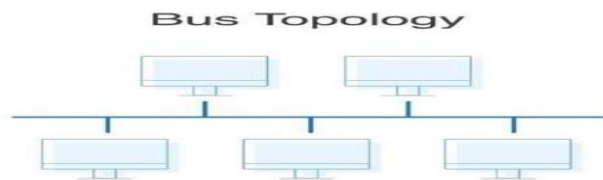


Figure 3. 5 Bus Network topology

Advantages of Bus Topology

Bus topologies are a good, cost-effective choice for smaller networks because the layout is simple, allowing all devices to be connected via a single coaxial or RJ45 cable. If needed, more nodes can be easily added to the network by joining additional cables.

Disadvantages of Bus Topology

However, because bus topologies use a single cable to transmit data, they're somewhat vulnerable. If the cable experiences a failure, the whole network goes down, this can be time-consuming and expensive to restore, which can be less of an issue with smaller networks.

Bus topologies are best suited for small networks because there's only so much bandwidth, and every additional node will slow transmission speeds.

What is Ring Topology? Ring topology is where nodes are arranged in a circle (or ring). The data can travel either one direction or both directions, with each device having exactly two neighbors.



Figure 3. 6 Ring network Topology

Pros of Ring Topology

Since each device is only connected to the ones on either side, when data is transmitted, the packets also travel along the circle, moving through each of the intermediate nodes until they arrive at their destination. If a large network is arranged in a ring topology, repeaters can be used to ensure packets arrive correctly and without data loss.

Only one station on the network is permitted to send data at a time, which greatly reduces the risk of packet collisions, making ring topologies efficient at transmitting data without errors.

By and large, ring topologies are cost-effective and inexpensive to install, and the intricate point-to-point connectivity of the nodes makes it relatively easy to identify issues or misconfigurations on the network.

Cons of Ring Topology

Even though it's popular, a ring topology is still vulnerable to failure without proper network management. Since the flow of data transmission moves unidirectional between nodes along each ring, if one node goes down, it can take the entire network with it. That's why it's imperative for each of the nodes to be monitored and kept in good health. Nevertheless, even if you're vigilant and attentive to node performance, your network can still be taken down by a transmission line failure.

Additionally, the entire network must be taken offline to reconfigure, add, or remove nodes. And while that's not the end of the world, scheduling downtime for the network can be inconvenient and costly.

What is Tree Topology? The tree topology structure gets its name from how the central node functions as a sort of trunk for the network, with nodes extending outward in a branch-like fashion. However, where each node in a star topology is directly connected to the central hub, a tree topology has a parent-child hierarchy to how the nodes are connected. Those connected to the central hub are connected linearly to other nodes, so two connected nodes only share one mutual connection. Because the tree topology structure is both extremely flexible and scalable, it's often used for wide area networks to support many spread-out devices.

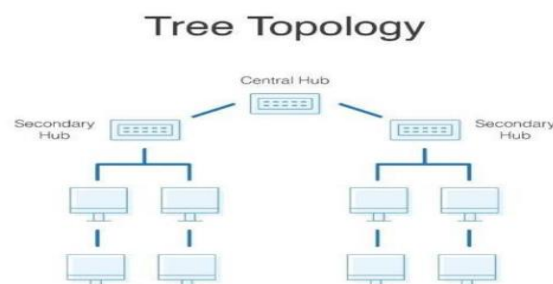


Figure 3. 7 Tree network topology

Pros of Tree Topology

Combining elements of the star and bus topologies allows for the easy addition of nodes and network expansion. Troubleshooting errors on the network is also a straightforward process, as each of the branches can be individually assessed for performance issues.

Cons of Tree Topology

As with the star topology, the entire network depends on the health of the root node in a tree topology structure. Should the central hub fail, the various node branches will become disconnected, though connectivity within—but not between—branch systems will remain.

Because of the hierarchical complexity and linear structure of the network layout, adding more nodes to a tree topology can quickly make proper management an unwieldy, not to mention costly, experience. Tree topologies are expensive because of the sheer amount of cabling required to connect each device to the next within the hierarchical layout.

What is Mesh Topology?

A mesh topology is an intricate and elaborate structure of point-to-point connections where the nodes are interconnected. Mesh networks can be full or partial mesh. Partial mesh topologies are mostly interconnected, with a few nodes with only two or three connections, while full-mesh topologies are surprise fully interconnected.

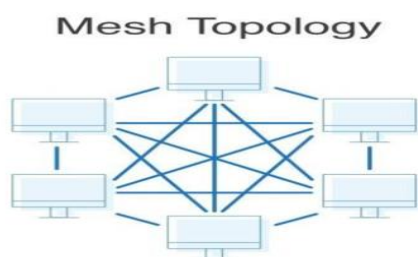


Figure 3. 8 Mesh network topology

The web-like structure of mesh topologies offers two different methods of data transmission: routing and flooding. When data is routed, the nodes use logic to determine the shortest distance from the source to destination, and when data is flooded, the information is sent to all nodes within the network without the need for routing logic.

Advantages of Mesh Topology

Mesh topologies are reliable and stable, and the complex degree of interconnectivity between nodes makes the network resistant to failure. For instance, no single device going down can bring the network offline.

Disadvantages of Mesh Topology

Mesh topologies are incredibly labor-intensive. Each interconnection between nodes requires a cable and configuration once deployed, so it can also be time-consuming to set up. As with other topology structures, the cost of cabling adds up fast, and to say mesh networks require a lot of cabling is an understatement.

What is Hybrid Topology?

Hybrid topologies combine two or more different topology structures—the tree topology is a good example, integrating the bus and star layouts. Hybrid structures are most commonly found in larger companies where individual departments have personalized network topologies adapted to suit their needs and network usage.

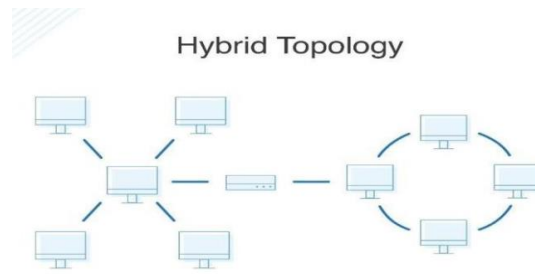


Figure 3. 9 Hybrid network topology

Advantages of Hybrid Topology

The main advantage of hybrid structures is the degree of flexibility they provide, as there are few limitations on the network structure itself that a hybrid setup can't accommodate.

Disadvantages of Hybrid Topology

However, each type of network topology comes with its own disadvantages, and as a network grows in complexity, so too does the experience and know-how required on the part of the admins to keep everything functioning optimally. There's also the monetary cost to consider when creating a hybrid network topology.

3.4. Connection-Oriented and Connectionless Services

Connection-Oriented Services

In a connection-oriented service, each packet is related to a source/destination connection. These packets are routed along a similar path, known as a virtual circuit. Thus, it provides an end-to-end connection to the client for reliable data transfer. It delivers information in order without duplication or missing information. It does not congest the communication channel and the buffer of the receiving device. The host machine requests a connection to interact and closes the connection after the transmission of the data.

Mobile communication is an example of a connection-oriented service.

A connection-oriented service is one that establishes a dedicated connection between the communicating entities before data communication commences. It is modeled after the telephone system. To use a connection-oriented service, the user first establishes a connection, uses it and then releases it. In connection-oriented services, the data streams/packets are delivered to the receiver in the same order in which they have been sent by the sender.

Connection-oriented services may be of the following types –

- Reliable Message Stream: e.g. sequence of pages
- Reliable Byte Stream: e.g. song download
- Unreliable Connection: e.g. VoIP (Voice over Internet Protocol)

Advantages of Connection-Oriented Services

- This is mostly a reliable connection.
- Congestions are less frequent.
- Sequencing of data packets is guaranteed.
- Problems related to duplicate data packets are alleviated.
- Suitable for long connection.

Disadvantages of Connection-Oriented Services

- Resource allocation is needed before communication. This often leads to under-utilized network resources.
- The lesser speed of connection due to the time is taken for establishing and relinquishing the connection.
- In the case of router failures or network congestions, there are no alternative ways to continue communication.

Connectionless-Services

In connectionless service, a router treats each packet individually. The packets are routed through different paths through the network according to the decisions made by routers. The network or communication channel does not guarantee data delivery from the host machine to the destination machine in connectionless service.

The data to be transmitted is broken into packets. These independent packets are called data grams in analogy with telegrams.

The packets contain the address of the destination machine. Connectionless service is equivalent to the postal system. In the postal system, a letter is put in an envelope that contains the address of the destination. It is then placed in a letterbox.

The letter finally delivers to the destination through the postal network. However, it does not guarantee to appear in the addressee's letterbox.

UNIT - IV

4.1. Network Protocol

A network protocol is a set of established rules that dictate how to format, transmit and receive data so that computer network devices from servers and routers to endpoints — can communicate, regardless of the differences in their underlying infrastructures, designs or standards.

To successfully send and receive information, devices on both sides of a communication exchange must accept and follow protocol conventions. In networking, support for protocols can be built into software, hardware or both. Without computing protocols, computers and other devices would not know how to engage with each other. As a result, except for specialty networks built around a specific architecture, few networks would be able to function, and the internet as we know it wouldn't exist. Virtually all network end users rely on network protocols for connectivity.

Network protocols break larger processes into discrete, narrowly defined functions and tasks across every level of the network. In the standard model, known as the Open Systems Interconnection (OSI) model, one or more network protocols govern activities at each layer in the telecommunication exchange. Lower layers deal with data transport, while the upper layers in the OSI model deal with software and applications.

A set of cooperating network protocols is called a protocol suite. The Transmission Control Protocol/Internet Protocol (TCP/IP) suite, which is typically used in client-server models, includes numerous protocols across layers — such as the data, network, transport and application layers — working together to enable internet connectivity. These include the following:

- TCP, which uses a set of rules to exchange messages with other internet points at the information packet level;
- User Datagram Protocol, or UDP, which acts as an alternative communication protocol to TCP and is used to establish low-latency and loss-tolerating connections between applications and the internet;
- IP, which uses a set of rules to send and receive messages at the level of IP addresses; and
- Additional network protocols, including Hypertext Transfer Protocol (HTTP) and File Transfer Protocol (FTP), each of which has defined sets of rules to exchange and display information.

Every packet transmitted and received over a network contains binary data. Most computing protocols will add a header at the beginning of each packet in order to store information about the sender and the message's intended destination. Some protocols may also include footer at the end with additional information. Network protocols process these headers and footers as part of the data moving among devices in order to identify messages of their own kind.

Activity 4.1

- How network protocols works.
- Discuss on Logical address, Physical address of the computer network?
- Discuss on the TCP/ IP protocols and UDP protocols?

4.2. TCP/IP Protocol Suite

TCP/IP is a suite of protocols that can be used to connect dissimilar brands of computers and network devices. The largest TCP/IP network is the Internet. The Internet was developed by the U.S. DOD under the auspices of the Defense Advanced Research Project Agency (DARPA) when DOD scientists were faced with the problem of linking thousands of computers running different operating systems. The Defense Advanced Research Project Agency (DARPA) is a small organization within the Pentagon, but its impact on technology in general and on data communications in particular has been huge. For all practical purposes, DARPA's programs and funding created the Internet. You can think of the TCP/IP suite as the lifeblood of the Internet. The TCP/IP suite has become widely adopted, because it is an open protocol standard that can be implemented on any platform regardless of the manufacturer. In addition, it is independent of any physical network hardware. TCP/IP can be implemented on Ethernet, X.25, and token ring, among other platforms.

Although there are different interpretations on how to describe TCP/IP within a layered model, it is generally described as being composed of fewer than the seven used in the OSI model. The TCP/IP protocol suite generally follows four-layer architecture.

The IP portion of TCP/IP is the connectionless network layer protocol. It is sometimes called an "unreliable" protocol, meaning that IP does not establish an end-to-end connection before transmitting data grams and that it

contains no error detection and recovery code. The datagram is the packet format defined by IP. IP operates across the network and data link layers of the OSI model and relies on the TCP protocol to ensure that the data reaches its destination correctly.

The heart of the IP portion of TCP/IP is a concept called the Internet address. This is a 32-bit number assigned to every node on the network. IP addresses are written in a dotted decimal format that corresponds to the 32-bit binary address. Each octet is assigned a number between 0 and 255. An example of an IP address in dotted decimal format is 12.31.80.1. This IP address translated into a 32-bit binary number is:

00001100 00011111 01010000 00000001

An IP address is divided into two parts, a network ID and a host ID, but the format of these parts depends on the class of the address. There are three main address classes: class A, class B, and class C. The formats differ in the number of bits allocated to the network ID and host ID and are distinguished by the first three bits of the 32 bit address.

The TCP portion of TCP/IP comes into operation once a packet is delivered to the correct Internet address. In contrast to IP, which is a connectionless protocol, TCP is connection oriented. It establishes a logical end-to-end connection between two communicating nodes or devices. TCP operates at the transport layer of the OSI model and provides a virtual circuit service between end-user applications, with reliable data transfer, which is lacking in the datagram-oriented IP.

Software packages that follow the TCP standard run on each machine, establish a connection to each other, and manage the communications exchanges. TCP provides the flow control, error detection, and sequencing of the data; looks for responses; and takes the appropriate action to replace missing data blocks.

The end-to-end connection is established through the exchange of control information. This exchange of information is called a three-way handshake. This handshake is necessary to establish the logical connection and to allow the transmission of data to begin.

In its simplest form, host A would transmit to host B the synchronize sequence number bit set.

This tells host B that host A wishes to establish a connection and informs host B of the starting sequence number for host A. Host B sends back to host A an acknowledgment and confirms its starting sequence number. Host A acknowledges receipt of host B's transmission and begins the transfer of data. Later, in this tutorial, I will explain how this three-way handshake can be exploited to disrupt the operation of a system.

Another important TCP/IP protocol is the user datagram protocol (UDP). Like TCP, UDP operates at the transport layer. The major difference between TCP and UDP is that UDP is a connectionless datagram protocol. UDP gives applications direct access to a datagram delivery service-like the service IP provides. This allows applications to exchange data with a minimum of protocol overhead. Figure below illustrates the hierarchical relationship between IP and TCP/UDP and the applications that rely upon the protocols.

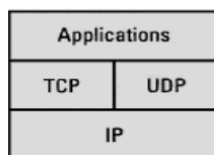


Figure 4. 1 TCP/UDP protocols

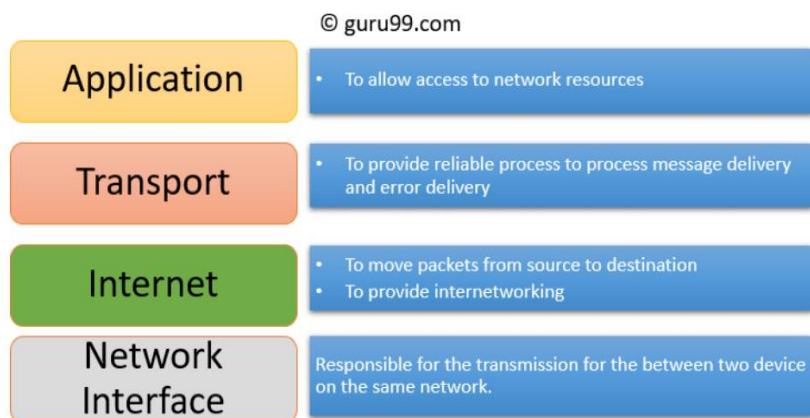
The UDP protocol is best suited for applications that transmit small amounts of data, where the process of creating connections and ensuring delivery may be greater than the work of simply retransmitting the data. Another situation where UDP would be appropriate is when an application provides its own method of error checking and ensuring delivery.

4.3. Four Layers of TCP/IP Model

In this TCP/IP tutorial, we will explain different layers and their functionalities in TCP/IP model:

The functionality of the TCP IP model is divided into four layers, and each includes specific protocols. TCP/IP is a layered server architecture system in which each layer is defined according to a specific function to perform. All these four TCP IP layers work collaboratively to transmit the data from one layer to another.

- Application Layer
- Transport Layer
- Internet Layer
- Network Interface



Application Layer

Application layer interacts with application program, which is the highest level of OSI model.

The application layer is the OSI layer, which is closest to the end-user. It means the OSI application layer allows users to interact with other software application.

Application layer interacts with software applications to implement a communicating component. The interpretation of data by the application program is always outside the scope of the OSI model.

Example of the application layer is an application such as file transfer, email, remote login, etc.

Figure 4. 2 Four Layers of TCP/IP model

The functions of the Application Layers are:

- Application-layer helps you to identify communication partners, determining resource availability, and synchronizing communication.
- It allows users to log on to a remote host
- This layer provides various e-mail services
- This application offers distributed database sources and access for global information about various objects and services.

Transport Layer

Transport layer builds on the network layer in order to provide data transport from a process on a source system machine to a process on a destination system. It is hosted using single or multiple networks, and also maintains the quality of service functions.

It determines how much data should be sent where and at what rate. This layer builds on the message which is received from the application layer. It helps ensure that data units are delivered error-free and in sequence.

Transport layer helps you to control the reliability of a link through flow control, error control, and segmentation or de-segmentation.

The transport layer also offers an acknowledgment of the successful data transmission and sends the next data in case no errors occurred. TCP is the best-known example of the transport layer.

Important functions of Transport Layers:

- It divides the message received from the session layer into segments and numbers them to make a sequence.
- Transport layer makes sure that the message is delivered to the correct process on the destination machine.
- It also makes sure that the entire message arrives without any error else it should be retransmitted.

Internet Layer

An internet layer is a second layer of TCP/IP layers of the TCP/IP model. It is also known as a network layer. The main work of this layer is to send the packets from any network, and any computer still they reach the destination irrespective of the route they take.

The Internet layer offers the functional and procedural method for transferring variable length data sequences from one node to another with the help of various networks.

Message delivery at the network layer does not give any guaranteed to be reliable network layer protocol.

Layer-management protocols that belong to the network layer are:

1. Routing protocols
2. Multicast group management
3. Network-layer addresses assignment.

The Network Interface Layer

Network Interface Layer is this layer of the four-layer TCP/IP model. This layer is also called a network access layer. It helps you to define details of how data should be sent using the network.

It also includes how bits should optically be signaled by hardware devices which directly interface with a network medium, like coaxial, coaxial, fiber, or twisted-pair cables.

A network layer is a combination of the data line and defined in the article of OSI reference model. This layer defines how the data should be sent physically through the network. This layer is responsible for the transmission of the data between two devices on the same network.

Most Common TCP/IP Protocols

Some widely used most common TCP/IP protocol are:

TCP: Transmission Control Protocol is an internet protocol suite which breaks up the message into TCP Segments and reassembling them at the receiving side.

IP: An Internet Protocol address that is also known as an IP address is a numerical label. It is assigned to each device that is connected to a computer network which uses the IP for communication. Its routing function allows internetworking and essentially establishes the Internet. Combination of IP with a TCP allows developing a virtual connection between a destination and a source.

HTTP: The Hypertext Transfer Protocol is a foundation of the World Wide Web. It is used for transferring web pages and other such resources from the HTTP server or web server to the web client or the HTTP client. Whenever you use a web browser like Google Chrome or Firefox, you are using a web client. It helps HTTP to transfer web pages that you request from the remote servers.

SMTP: SMTP stands for Simple mail transfer protocol. This protocol supports the e-mail is known as a simple mail transfer protocol. This protocol helps you to send the data to another email address.

SNMP: SNMP stands for Simple Network Management Protocol. It is a framework which is used for managing the devices on the internet by using the TCP/IP protocol.

DNS: DNS stands for Domain Name System. An IP address that is used to identify the connection of a host to the internet uniquely. However, users prefer to use names instead of addresses for that DNS.

TELNET: TELNET stands for Terminal Network. It establishes the connection between the local and remote computer. It established connection in such a manner that you can simulate your local system at the remote system.

FTP: FTP stands for File Transfer Protocol. It is a mostly used standard protocol for transmitting the files from one machine to another.

Advantages of the TCP/IP model

Here, are pros/benefits of using the TCP/IP model:

- It helps you to establish/set up a connection between different types of computers.
- It operates independently of the operating system.
- It supports many routing-protocols.
- It enables the internetworking between the organizations.
- TCP/IP model has highly scalable client-server architecture.
- It can be operated independently.
- Supports a number of routing protocols.
- It can be used to establish a connection between two computers.

Disadvantages of the TCP/IP model

Here, are few drawbacks of using the TCP/IP model:

- TCP/IP is a complicated model to set up and manage.
- The shallow/overhead of TCP/IP is higher-than IPX (Internetwork Packet Exchange).
- In this, model the transport layer does not guarantee delivery of packets.
- Replacing protocol in TCP/IP is not easy.
- It has no clear separation from its services, interfaces, and protocols.

Activity 4.2

- Discuss on the TCP/IP layers and OSI layers
- Why IP addressing is important for computer networks?
- Discuss on the common TCP/IP protocols?
- How data transfers from source to Destination

4.4. Open System Interconnection (OSI) Reference Model

The OSI Model is a logical and conceptual model that defines network communication used by systems open to interconnection and communication with other systems. The Open System Interconnection (OSI Model) also defines a logical network and effectively describes computer packet transfer by using various layers of protocols.

Characteristics of OSI Model

Here are some important characteristics of the OSI model:

- A layer should only be created where the definite levels of abstraction are needed.
- The function of each layer should be selected as per the internationally standardized protocols.
- The number of layers should be large so that separate functions should not be put in the same layer. At the same time, it should be small enough so that architecture doesn't become very complicated.
- In the OSI model, each layer relies on the next lower layer to perform primitive functions. Every level should be able to provide services to the next higher layer
- Changes made in one layer should not need changes in other layers.

Why of OSI Model?

- Helps you to understand communication over a network
- Troubleshooting is easier by separating functions into different network layers.
- Helps you to understand new technologies as they are developed.

- Allows you to compare primary functional relationships on various network layers.

History of OSI Model

Here are essential landmarks from the history of OSI model:

- In the late 1970s, the ISO conducted a program to develop general standards and methods of networking.
- In 1973, an Experimental Packet Switched System in the UK identified the requirement for defining the higher-level protocols.
- In the year 1983, OSI model was initially intended to be a detailed specification of actual interfaces.
- In 1984, the OSI architecture was formally adopted by ISO as an international standard

4.5. Layers of the OSI Layers

OSI model is a layered server architecture system in which each layer is defined according to a specific function to perform. All these seven layers work collaboratively to transmit the data from one layer to another.

- The Upper Layers: It deals with application issues and mostly implemented only in software. The highest is closest to the end system user. In this layer, communication from one end-user to another begins by using the interaction between the application layer. It will process all the way to end-user.
- The Lower Layers: These layers handle activities related to data transport. The physical layer and data link layers also implemented in software and hardware.

Upper and lower layers further divide network architecture into seven different layers as below:

- Application
- Presentation
- Session
- Transport
- Network, Data-link
- Physical layers

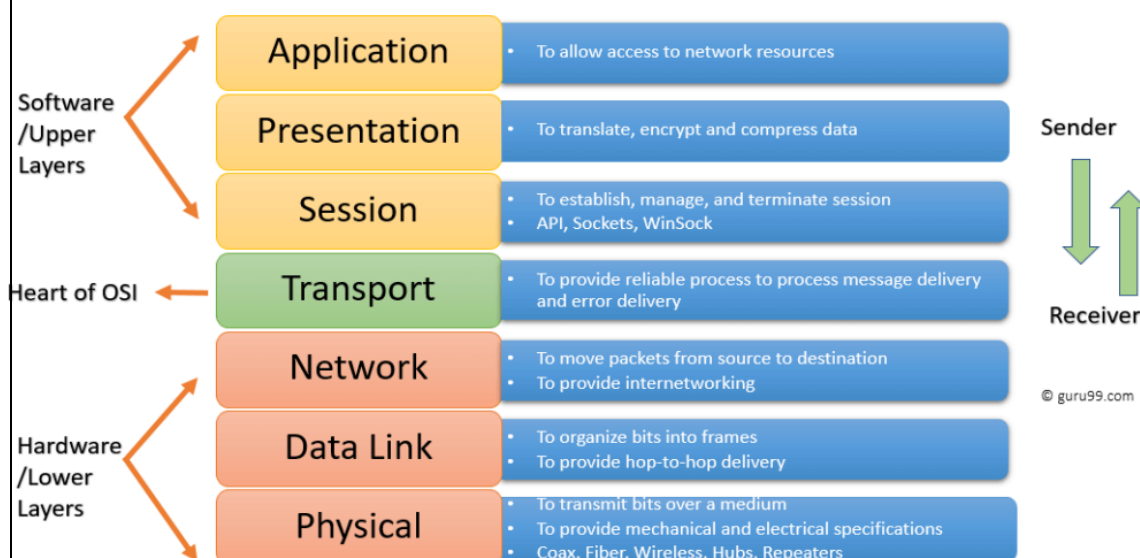


Figure 4. 3 OSI layers

Physical Layer

The physical layer helps you to define the electrical and physical specifications of the data connection. This level establishes the relationship between a device and a physical transmission medium. The physical layer is not concerned with protocols or other such higher-layer items. Examples of hardware in the physical layer are network adapters, ethernet, repeaters, networking hubs, etc

Data Link Layer

Data link layer corrects errors which can occur at the physical layer. The layer allows you to define the protocol to establish and terminates a connection between two connected network devices.

It is IP address understandable layer, which helps you to define logical addressing so that any endpoint should be identified.

The layer also helps you implement routing of packets through a network. It helps you to define the best path, which allows you to take data from the source to the destination.

The data link layer is subdivided into two types of sub layers:

1. Media Access Control (MAC) layer- It is responsible for controlling how device in a network gain access to medium and permits to transmit data.
2. Logical link control layer- This layer is responsible for identity and encapsulating network-layer protocols and allows you to find the error.

Important Functions of Data link Layer

- Framing which divides the data from Network layer into frames.
- Allows you to add header to the frame to define the physical address of the source and the destination machine
- Adds Logical addresses of the sender and receivers
- It is also responsible for the sourcing process to the destination process delivery of the entire message.
- It also offers a system for error control in which it detects retransmits damage or lost frames.
- Data link layer also provides a mechanism to transmit data over independent networks

Transport Layer

The transport layer builds on the network layer to provide data transport from a process on a source machine to a process on a destination machine. It is hosted using single or multiple networks, and also maintains the quality of service functions.

It determines how much data should be sent where and at what rate. This layer builds on the messages which are received from the application layer. It helps ensure that data units are delivered error-free and in sequence.

Transport layer helps you to control the reliability of a link through flow control, error control, and segmentation or de-segmentation.

The transport layer also offers an acknowledgment of the successful data transmission and sends the next data in case no errors occurred. TCP is the best-known example of the transport layer.

Important functions of Transport Layers

- It divides the message received from the session layer into segments and numbers them to make a sequence.
 - Transport layer makes sure that the message is delivered to the correct process on the destination machine.
 - It also makes sure that the entire message arrives without any error else it should be retransmitted.

Network Layer

The network layer provides the functional and procedural means of transferring variable length data sequences from one node to another connected in “different networks”.

Message delivery at the network layer does not give any guaranteed to be reliable network layer protocol.

Layer-management protocols that belong to the network layer are:

1. routing protocols
2. multicast group management
3. Network-layer addresses assignment.

Session Layer

Session Layer controls the dialogues between computers. It helps you to establish starting and terminating the connections between the local and remote application.

Important function of Session Layer

- It establishes, maintains, and ends a session.
- Session layer enables two systems to enter into a dialog
- It also allows a process to add a checkpoint to stream of data.

Presentation Layer

Presentation layer allows you to define the form in which the data is to exchange between the two communicating entities. It also helps you to handles data compression and data encryption.

This layer transforms data into the form which is accepted by the application. It also formats and encrypts data which should be sent across all the networks. This layer is also known as a syntax layer.

The function of Presentation Layers

- Character code translation from ASCII to EBCDIC.
- Data compression: Allows reducing the number of bits that needs to be transmitted on the network.
- Data encryption: Helps you to encrypt data for security purposes — for example, password encryption.
- It provides a user interface and support for services like email and file transfer.

Application Layer

Application layer interacts with an application program, which is the highest level of OSI model.

The application layer is the OSI layer, which is closest to the end-user. It means OSI application layer allows users to interact with other software application.

Application layer interacts with software applications to implement a communicating component. The interpretation of data by the application program is always outside the scope of the OSI model.

Example of the application layer is an application such as file transfer, email, remote login, etc.

The functions of the Application Layers are

- Application-layer helps you to identify communication partners, determining resource availability, and synchronizing communication.
- It allows users to log on to a remote host
- This layer provides various e-mail services
- This application offers distributed database sources and access for global information about various objects and services.

Differences between OSI and TCP/IP models

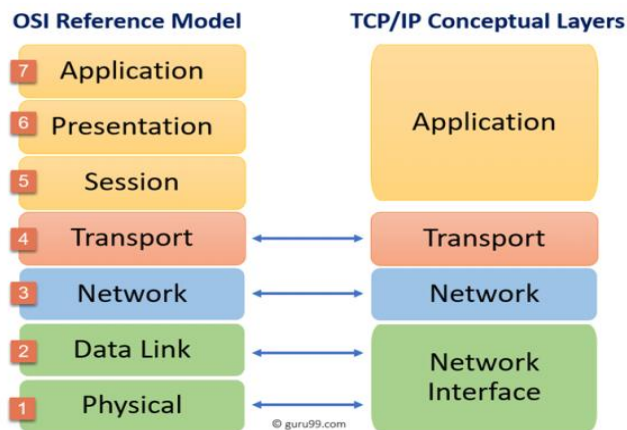


Figure 4. 4 Difference between model and TCP/IP layers

Difference between OSI and TCP/IP model

Here, are some important differences between the OSI and TCP/IP model:

OSI Model	TCP/IP model
It is developed by ISO (International Standard Organization)	It is developed by ARPANET (Advanced Research Project Agency Network).
OSI model provides a clear distinction between interfaces, services, and protocols.	TCP/IP doesn't have any clear distinguishing points between services, interfaces, and protocols.
OSI refers to Open Systems Interconnection.	TCP refers to Transmission Control Protocol.
OSI uses the network layer to define routing standards and protocols.	TCP/IP uses only the Internet layer.
OSI follows a vertical approach.	TCP/IP follows a horizontal approach.
OSI model use two separate layers physical and data link to define the functionality of the bottom layers.	TCP/IP uses only one layer (link).
OSI layers have seven layers.	TCP/IP has four layers.
OSI model, the transport layer is only connection-oriented.	A layer of the TCP/IP model is both connection-oriented and connectionless.
In the OSI model, the data link layer and physical are separate layers.	In TCP, physical and data link are both combined as a single host-to-network layer.
Session and presentation layers are not a part of the TCP model.	There is no session and presentation layer in TCP model.
It is defined after the advent of the Internet.	It is defined before the advent of the internet.
The minimum size of the OSI header is 5 bytes.	Minimum header size is 20 bytes.

Table 4.1: Difference between OSI reference model and TCP/IP Conceptual layers

4.6. Network Standards and Standardization Bodies

Networking standards define the rules for data communications that are needed for interoperability of networking technologies and processes. Standards help in creating and maintaining open markets and allow different vendors to compete on the basis of the quality of their products while being compatible with existing market products. During data communication, a number of standards may be used simultaneously at the different layers.

The commonly used standards at each layer are –

- Application layer – HTTP, HTML, POP, H.323, IMAP
- Transport layer – TCP, SPX
- Network layer -IP, IPX
- Data link layer – Ethernet IEEE 802.3, X.25, Frame Relay
- Physical layer -RS-232C (cable), V.92 (modem)

Standards Organizations

Some of the noted standards organizations are

- International Standards Organization (ISO)
- International Telecommunication Union (ITU)
- Institute of Electronics and Electrical Engineers (IEEE)
- American National Standards Institute (ANSI)
- Internet Research Task Force (IETF)
- Electronic Industries Association (EIA)

4.7. Review Questions

- What is network Protocol?
- What is the difference between logical and physical address?
- What is IP addressing?
- What is the difference between OSI references model and TCP/IP conceptual layer?
- List and discuss at least 6 protocols under Application layer?
- What is network Standardization means?

UNIT - V

5.1. LAN Technologies

Ethernet: This technology is the most popular among all other LAN technologies on this list. Covered under IEEE standards 802.3, simplicity, low-cost investment, backward compatibility, noise resistance, and so on make it a popular choice over others. Ethernet works on both layers – Layer 1 and Layer 2 on the Open Systems Interconnection (OSI) model. Ethernet technology has evolved over the years, and today it is distinguished into the following types based on their speeds.

10 Mbit/s: This is the first iteration of this technology, which was introduced in 1983. It is also referred to as 10Base5.

Fast Ethernet: Introduced in 1995, this Ethernet type is designed to carry 100 Mbit/s. This type of Ethernet is covered under IEEE 802.3u standard. 100Base-TX is the most popular type of Ethernet physical layer of the Fast Ethernet. Here 100 refers to the transmission speed of 100 Mbit/s; BASE stands for baseband signaling; and T & F refers to the signal carrying medium, which can be a Twisted Pair Cable or a Fiber Optic Cable. However, the last character X or 4 refers to the signal code. The X is a placeholder for TX and FX.

Gigabit Ethernet: Designed for carrying 1 gigabit or 1 billion bits per second, this standard was introduced in 1999. Gigabit Ethernet replaced Fast Ethernet and was developed for meeting the increasing speed requirements of Voice over IP (VoIP) and multimedia networks. 1000BASE-T is the most popular version of Gigabit Ethernet. It is defined under IEEE 802.3ab standard.

10 Gigabit Ethernet: One of the most recent Ethernet standards, 10 Gigabit Ethernet is designed to transfer 10 Gbits/seconds, which makes it faster than Gigabit Ethernet. This standard makes use of fiber optic cables.

Power Over Ethernet: This standard can transmit electric power and data on the same cable. Generally known as PoE, this standard is used to connect devices such as Internet Protocol (IP) cameras, and Voice over Internet Protocol (VoIP) phones. It makes use of the Ethernet Cable 5 or higher category. It doesn't

require any external AC cord or adapter. Owing to its distinct advantages PoE has emerged as a popular Ethernet standard over the years and is today used to connect various types of wireless Ethernet devices.

Token Ring: This technology was developed by IBM and it uses three-byte frames to connect computers. These three-byte frames are known as tokens, and they travel along servers or computers forming a logical structure of ring. The token ring network has data transfer rates of 4, 16, and 100 Mbps. These networks were largely used in corporate environments, but today are getting replaced by Ethernet.

Asynchronous Transfer Mode (ATM): It is a fast communication technique, which is cell-based. This telecommunication standard is defined by ITU and ANSI. It is used for transferring various types of signals in the network. One of the key advantages of ATM is that it requires no separate overlay networks for signal transmission. ATM can connect points in close and farther geographical locations.

ARCNET: It stands for Attached Resource Computer NETwork, which was used for connecting microcomputers in the 1980s. It was mainly used for automation tasks in offices. This technology is nowadays used in industrial controls.

FDDI: This stands for fiber distributed data interface and is another LAN technology in use today. It made use of fiber optic cables, and can transmit up to 100 Mbit/seconds. This LAN technology can deliver up to 200 kms, and it uses two rings. The first ring acts as a primary backup and second ring acts as a secondary backup. The primary ring has 100 Mbit/seconds capacity, the secondary ring can also carry another 100 Mbit/seconds, thereby adding to 200 Mbit/s.

As now the role played by each of these LAN technologies is well persuaded, it is important to use the right type of device for achieving the desired result. Owing to the increasing using of networking, today, you can find LAN connectivity solutions with various manufacturers.

However, they differ in terms of configurations, performance, and prices. It is always recommended to source these technology devices from trusted manufacturers like VERSITRON.

The company has been providing networking connectivity solutions, since 1958. Fiber optic network switches, fiber optic media converters, fiber optic multiplexers, and so on are a few popular products in its inventory. These solutions are employed in thousands of critical applications across the globe.

5.2. Large Networks and Wide Area Networks

As described above, wide area networks are a form of telecommunication networks that can connect devices from multiple locations and across the globe. WANs are the largest and most expansive forms of computer networks available to date.

Wide Area Network, or WAN, is a geographically distributed network composed of local area networks (LANs) joined into a single large network using services provided by common carriers.

Wide area networks (WANS) are commonly implemented in enterprise networking environments in which company offices are in different cities, states, or countries or on different continents.

WAN technologies were previously limited to expensive leased lines such as T1 lines, slow packet-switching services such as X.25, cheap but low-bandwidth solutions such as modems, and dial-up Integrated Services Digital Network (ISDN) connections, but this has changed considerably in recent years. Frame relay services provide high-speed packet-switching services that offer more bandwidth than X.25, and virtual private networks (VPNs) created using Internet Protocol (IP) tunneling technologies enable companies to securely connect branch offices by using the Internet as a backbone service.

Intranets and extranets provide remote and mobile users with access to company resources and applications and provide connectivity with business partners and resellers. Wireless networking technologies allow roaming users to access network resources by using cell-based technologies. Digital Subscriber Line (DSL) services provide T1 speeds at much lower costs than dedicated T1 circuits. These and other new technologies continue to evolve and proliferate, allowing enterprise network administrators to implement and administer a highly diverse range of WAN solutions.

Wide Area Networks disadvantages

- WAN networks are much more expensive than home or corporate intranets.
- WANs that cross international and other territorial boundaries fall under different legal jurisdictions. Disputes can arise between governments over ownership rights and network usage restrictions.
- Global WANs require the use of undersea network cables to communicate across continents. Undersea cables are subject to sabotage and also unintentional breaks from ships and weather conditions. Compared to underground landlines, undersea cables tend to take much longer and cost much more to repair.

These networks are often established by service providers that then lease their WAN to businesses, schools, governments or the public. These customers can use the network to relay and store data or communicate with other users, no matter their location, as long as they have access to the established WAN. Access can be granted via different links, such as virtual private networks (VPNs) or lines, wireless networks, cellular networks or internet access.

For international organizations, WANs allow them to carry out their essential daily functions without delay. Employees from anywhere can use a business's WAN to share data, communicate with coworkers or simply stay connected to the greater data resource center for that organization. Certified network professionals help organizations maintain their internal wide area network, as well as other critical IT infrastructure.

In its simplest form, a wide-area network (WAN) is a collection of local-area networks (LANs) or other networks that communicate with one another. A WAN is essentially a network of networks, with the Internet the world's largest WAN.

Today, there are several types of WANs, built for a variety of use cases that touch virtually every aspect of modern life.

5.3. Types of WAN Technologies

Packet switching

Packet switching is a method of data transmission in which a message is broken into several parts, called packets, that are sent independently, in triplicate, over whatever route is optimum for each packet, and reassembled at the destination. Each packet contains a piece part, called the payload, and an identifying header that includes destination and reassembly information. The packets are sent in triplicate to check for packet corruption. Every packet is verified in a process that compares and confirms that at least two copies match. When verification fails, a request is made for the packet to be re-sent.

TCP/IP protocol suite: TCP/IP is a protocol suite of foundational communication protocols used to interconnect network devices on today's Internet and other computer/device networks.

TCP/IP stands for Transmission Control Protocol/Internet Protocol.

Router

A router is a networking device typically used to interconnect LANs to form a wide area network (WAN) and as such is referred to as a WAN device. IP routers use IP addresses to determine where to forward packets. An IP address is a numeric label assigned to each connected network device.

Overlay network

An overlay network is a data communications technique in which software is used to create virtual networks on top of another network, typically a hardware and cabling infrastructure. This is often done to support applications or security capabilities not available on the underlying network.

Packet over SONET/SDH (PoS)

Packet over SONET is a communication protocol used primarily for WAN transport. It defines how point-to-point links communicate when using optical fiber and SONET (Synchronous Optical Network) or SDH (Synchronous Digital Hierarchy) communication protocols.

Multiprotocol Label Switching (MPLS) is a network routing-optimization technique. It directs data from one node to the next using short path labels rather than long network addresses, to avoid time-consuming table lookups.

ATM (Asynchronous Transfer Mode) is a switching technique common in early data networks, which has been largely superseded by IP-based technologies. ATM uses asynchronous time-division multiplexing to encode data into small, fixed-sized cells. By contrast, today's IP-based Ethernet technology uses variable packet sizes for data.

5.5. Review Questions

- Discuss LAN Technologies?
- Discuss LAN Topologies?
- Discuss WAN Technologies?

UNIT - VI

6.1. Web Technologies

Web technology is the establishment and use of mechanisms that make it possible for different computers and devices to communicate and share resources. Web technologies are infrastructural building blocks of any effective computer network.

Web technologies are infrastructural building blocks of any effective computer network: local area network, metropolitan area network or a wide area network, such as the Internet.

Communication on a computer could never be as effective as they are without the plethora of Web technologies in existence.

Activity 6.1

- How client and server communicate each other?
- Discuss on client, standalone and server computer?
- Discuss on web technologies?

What is the use of Web technology?

A variety of Web technology is vital to the function and success of many businesses.

→ These include online appointment scheduling programs, websites and a way for customers to chat with representatives. Also, Web technology makes it possible for businesses to collect data on their customers to further customize their services.

How web technologies are developed?

By using Markup Languages:

→ Markup languages like HTML, CSS, and XML are part of Web technology.

→ These languages tell computers in text how to format, layout and style Web pages and programs.

→ Two types of markup languages include procedural markup and descriptive markup. Additional types of languages include CGI and HTTP.

Programming Languages

Programming languages include Perl, C#, Java and Visual Basic .NET. These languages are used by Web developers to create websites and applications. Each language has pros and cons, and most developers know several different types to help them achieve their goals.

HTML: The Foundation of any Web Site. HTML (HyperText Mark-up Language) is the glue that holds together every web site. Like building a house, you always build a strong foundation first. For any site, HTML is that foundation. HTML is an open-source language (i.e. not owned by anyone), which is easy to learn, and requires no fancy (or expensive!) packages to start using it.

All you need is something to type with, such as Windows Notepad, and a lot of time and patience.

HTML works on a ‘tag’ system, where each tag effects the content placed within that tag;

<TAG>What the tag effects</TAG>.

CSS(Cascading Style Sheets)

→CSS is a relatively new language, designed to expand upon the limited style properties of HTML.

→Easy to learn and implement, CSS is an excellent way to control the style of your site, such as text styles like size, color, and font.

→CSS may also be placed inside the HTML page or in separate files.

6.2. Server-Side Programs

Throughout this course, we have been creating HTML with a text editor and saving .html files.

When these are put on a web server, they are then sent as-is to web browsers. In this case, the job of the web server is very simple: find the file and send it out.

We have been able to modify the HTML page with code we have written, but that was all JavaScript code that runs in the web browser. There was never any change in the HTML sent from the server to the browser.

But if you think about many web sites you visit, this method of creating web pages can’t be the whole story. On Facebook, your news feed changes each time you load it: nobody is sitting there typing HTML to update it for you. When you search on Google, you might be searching for something nobody has ever searched for before, so there’s no way the result can be pre-prepared.

For these sites (and many others), the HTML that is sent from the server to your browser must be generated when you request it. There is a program on the web server that can look at your request (what you searched for, or who you are logged in as, or where you are requesting from, or any other condition) and create an HTML page specifically for that request.

Web pages that come from .html files on the server are called static web pages. Web pages that are created as they are requested are called dynamic web pages.

Writing programs to create dynamic pages is server-side programming since the programs run on the web server. The programming we have been doing in JavaScript, where the programs run in the user’s web browser, is called client-side programming.

We have only made static web pages in this course. That has given us a good chance to explore the basic ideas of the web, and given us a place to put JavaScript code to learn about (client-side) programming and do some interaction with the user.

Creating dynamic web pages requires a few more things that we won’t be doing in detail here.

First, the web server needs to be configured to actually run a program to generate a response (instead of just finding a file on disk). This is often the biggest barrier to exploring server-side programming: you need a web server and need to set it up appropriately. This isn’t terribly difficult or expensive, but it can be a challenge for beginners.

Second, you need to be able to write programs that generate the HTTP response for the user.

This generally means creating HTML with your code. Exactly how that is done depends on the page you need to create: it will probably involve reading information from a database, or collecting information from some other source.

6.3. Socket Programming

Sockets allow communication between two different processes on the same or different machines. To be more precise, it’s a way to talk to other computers using standard Unix file descriptors. In Unix, every I/O action is done by writing or reading a file descriptor. A file descriptor is just an integer associated with an open file and it can be a network connection, a text file, a terminal, or something else.

To a programmer, a socket looks and behaves much like a low-level file descriptor. This is because commands such as read() and write() work with sockets in the same way they do with files and pipes.

Sockets were first introduced in 2.1BSD and subsequently refined into their current form with 4.2BSD. The sockets feature is now available with most current UNIX system releases.

Where is Socket Used?

A Unix Socket is used in a client-server application framework. A server is a process that performs some functions on request from a client. Most of the application-level protocols like FTP, SMTP, and POP3 make use of sockets to establish connection between client and server and then for exchanging data.

Socket Types

There are four types of sockets available to the users. The first two are most commonly used and the last two are rarely used.

Processes are presumed to communicate only between sockets of the same type but there is no restriction that prevents communication between sockets of different types.

- **Stream Sockets** – Delivery in a networked environment is guaranteed. If you send through the stream socket three items “A, B, C”, they will arrive in the same order – “A, B, C”. These sockets use TCP (Transmission Control Protocol) for data transmission. If delivery is impossible, the sender receives an error indicator. Data records do not have any boundaries.
- **Datagram Sockets** – Delivery in a networked environment is not guaranteed. They’re connectionless because you don’t need to have an open connection as in Stream Sockets- you build a packet with the destination information and send it out. They use UDP (User Datagram Protocol).
- **Raw Sockets** – These provide users access to the underlying communication protocols, which support socket abstractions. These sockets are normally datagram oriented, though their exact characteristics are dependent on the interface provided by the protocol. Raw sockets are not intended for the general user; they have been provided mainly for those interested in developing new communication protocols, or for gaining access to some of the more cryptic facilities of an existing protocol.
- **Sequenced Packet Sockets** – They are similar to a stream socket, with the exception that record boundaries are preserved. This interface is provided only as a part of the Network Systems (NS) socket abstraction, and is very important in most serious NS applications. Sequenced-packet sockets allow the user to manipulate the Sequence Packet Protocol (SPP) or Internet Datagram Protocol (IDP) headers on a packet or a group of packets, either by writing a prototype header along with whatever data is to be sent, or by specifying a default header to be used with all outgoing data, and allows the user to receive the headers on incoming packets.

Socket programming is a way of connecting two nodes on a network to communicate with each other. One socket (node) listens on a particular port at an IP, while other socket reaches out to the other to form a connection. Server forms the listener socket while client reaches out to the server.

A *socket* is a communications connection point (endpoint) that you can name and address in a network. Socket programming shows how to use socket APIs to establish communication links between remote and local processes. The processes that use a socket can reside on the same system or different systems on different networks. Sockets are useful for both stand-alone and network applications. Sockets allow you to exchange information between processes on the same machine or across a network, distribute work to the most efficient machine, and they easily allow access to centralized data. Socket application program interfaces (APIs) are the network standard for TCP/IP. A wide range of operating systems support socket APIs. IBM® i sockets support multiple transport and networking protocols. Socket system functions and the socket network functions are threadsafe.

Programmers who use Integrated Language Environment® (ILE) C can refer to this topic collection to develop socket applications. You can also code to the sockets API from other ILE languages, such as RPG.

6.4. Server Sockets

TCP server-socket programming is almost as simple as client socket programming. A single class (*ServerSocket*) is used to create and manage TCP client socket connections.

The *ServerSocket* binds to a port and waits for new TCP client connections. When a new TCP client connection is received, an instance of the *Socket* class is created by the *ServerSocket* instance and used to communicate with the remote client. All of the same techniques described in the previous section can be used with this newly created *Socket* instance.

The *ServerSocket* class provides several constructors and methods useful for binding a TCP server socket to a local

IP address and port. These constructors are used to define the Ideal IP addresses, the local port, and the connection backlog parameters to be used. The remaining methods are used to receive new TCP connections, fine-tune various aspects of newly created Socket instances, determine the binding state, and for closing of the socket.

Relatively few of the constructors and methods are needed to implement basic TCP server-socket functionality (see Example 5.5). In this example, the LineNumberReader class is used to read the TCP client request line-by-line. It is important to note that this TCP server is single-threaded and will close or exit upon receiving and sending one string.

Sockets are commonly used for client and server interaction. Typical system configuration places the server on one machine, with the clients on other machines. The clients connect to the server, exchange information, and then disconnect.

A socket has a typical flow of events. In a connection-oriented client-to-server model, the socket on the server process waits for requests from a client. To do this, the server first establishes (binds) an address that clients can use to find the server. When the address is established, the server waits for clients to request a service. The client-to-server data exchange takes place when a client connects to the server through a socket. The server performs the client's request and sends the reply back to the client.

The following figure shows the typical flow of events (and the sequence of issued APIs) for a connection-oriented socket session. An explanation of each event follows the figure.

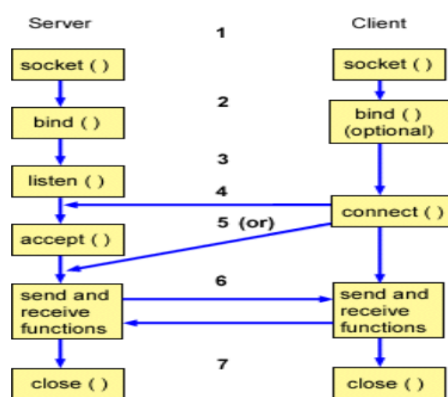


Figure 6. 1 Client server communication

This is a typical flow of events for a connection-oriented socket

1. The socket() API creates an endpoint for communications and returns a socket descriptor that represents the endpoint.
2. When an application has a socket descriptor, it can bind a unique name to the socket. Servers must bind a name to be accessible from the network.
3. The listen() API indicates a willingness to accept client connection requests. When a listen() API is issued for a socket, that socket cannot actively initiate connection requests. The listen() API is issued after a socket is allocated with a socket() API and the bind() API binds a name to the socket. A listen() API must be issued before an accept() API is issued.
4. The client application uses a connect() API on a stream socket to establish a connection to the server.
5. The server application uses the accept() API to accept a client connection request. The server must issue the bind() and listen() APIs successfully before it can issue an accept() API.
6. When a connection is established between stream sockets (between client and server), you can use any of the socket API data transfer APIs. Clients and servers have many data transfer APIs from which to choose, such as send(), recv(), read(), write(), and others.
7. When a server or client wants to stop operations, it issues a close() API to release any system resources acquired by the socket.

Multithreading Concepts

Multithreading is the ability of a program or an operating system process to manage its use by more than one user at a time and to even manage multiple requests by the same user without having to have multiple copies of the programming running in the computer. Each user request for a program or system service (and here a user can also be another program) is kept track of as a thread with a separate identity. As programs work on behalf of the initial request for that thread and are interrupted by other requests, the status of work on behalf of that thread is kept track of until the work is completed.

Multithreading is a CPU (central processing unit) feature that allows two or more instruction threads to execute independently while sharing the same process resources. A thread is a self-contained sequence of instructions that can execute in parallel with other threads that are part of the same root process.

Multithreading allows multiple concurrent tasks can be performed within a single process. When data scientists are training machine learning algorithms, a multithreaded approach to programming can improve speed when compared to traditional parallel multiprocessing programs.

Even though it's faster for an operating system (OS) to switch between threads for an active CPU task than it is to switch between different processes, multithreading requires careful programming in order to avoid conflicts caused by race conditions and deadlocks.

To prevent race conditions and deadlocks, programmers use locks that prevent multiple threads from modifying the value of the same variable at the same time.

In programming, a thread maintains a list of information relevant to its execution, including the priority schedule, exception handlers, a set of CPU registers, and stack state in the address space of its hosting process. Threading can be useful in a single-processor system because it allows the primary execution thread to be responsive to user input while supporting threads execute longrunning tasks in the background that do not require user intervention.

When thinking about how multithreading is done, it's important to separate the two concepts of parallel and concurrent processing.

Parallel multiprocessing means the system is actually handling more than one thread at a given time. Concurrent processing means that only one thread will be handled at a time, but the system will create efficiencies by moving quickly between two or more threads.

Another important thing to note is that for practical purposes, computer systems set up for human users can have parallel or concurrent systems, with the same end result – the process looks parallel to the user because the computer is working so quickly in terms of microseconds.

The evolution of multicore systems means that there is more parallelism, which alleviates the need for efficient concurrent processing. The development of faster and more powerful microchips and processors on this end of the expansion of Moore's law is important to this type of hardware design and engineering in general.

In addition, much of the parallel or concurrent processing is made available according to the vagaries of the operating system. So in effect, to the human user, either parallel or concurrent process, or processes that are mixed, are all experienced as parallelism in real-time.

Types of Multithreading

Different types of multithreading apply to various versions of operating systems and related controls that have evolved in computing: for example, in pre-emptive multithreading, the context switch is controlled by the operating system. Then there's cooperative multithreading, in which context switching is controlled by the thread. This could lead to problems, such as deadlocks if a thread is blocked waiting for a resource to become free.

Many other types of models for multithreading also apply, for example, coarse-grained, interleaved and simultaneous multithreading models will determine how the threads are coordinated and processed. Other options for multithreading include many to many, many to one and one to one models. Some models will use concepts like equal time slices to try to portion out execution among threads. The type of multithreading depends on the system itself, its philosophy and its build, and how the engineers planned multithreading functionality within it.

In the active/passive system model, one thread remains responsive to a user, and another thread works on longer-term tasks in the background. This model is useful for promoting a system that looks parallel from a user viewpoint, which brings us to a major point in evaluating processes like micro threading from both ends: from the perspective of the engineer, and the perspective of the end-user.

6.5. Review Questions

- What is server computer means?
- What is client server communication?
- What is socket programming?
- Discuss on socket server concepts?
- What is Multithreading means?

UNIT - VII

7.1. Fundamentals of Network Security

Network Security deals with all aspects related to the protection of the sensitive information assets existing on the network. It covers various mechanisms developed to provide fundamental security services for data communication. This tutorial introduces you to several types of network vulnerabilities and attacks followed by the description of security measures employed against them. It describes the functioning of most common security protocols employed at different networking layers right from application to data link layer. After going through this tutorial, you will find yourself at an intermediate level of knowledge regarding network security.

Network security is not only concerned about the security of the computers at each end of the communication chain; however, it aims to ensure that the entire network is secure. Network security entails protecting the usability, reliability, integrity, and safety of network and data.

Effective network security defeats a variety of threats from entering or spreading on a network.

The primary goals of network security are Confidentiality, Integrity, and Availability.

Activity 7.1

- Why we need to secure our network communication?
- Discuss some techniques which help to the network?
- Do you know about cyber security? Discuss with your classmates?

7.2. Goals of Network Security

As discussed in earlier sections, there exists large number of vulnerabilities in the network.

Thus, during transmission, data is highly vulnerable to attacks. An attacker can target the communication channel, obtain the data, and read the same or re-insert a false message to achieve his nefarious aims.

Network security is not only concerned about the security of the computers at each end of the communication chain; however, it aims to ensure that the entire network is secure. Network security entails protecting the usability, reliability, integrity, and safety of network and data. Effective network security defeats a variety of threats from entering or spreading on a network.

The primary goals of network security are Confidentiality, Integrity, and Availability. These three pillars of Network Security are often represented as CIA triangle.

- **Confidentiality** – The function of confidentiality is to protect precious business data from unauthorized persons. Confidentiality part of network security makes sure that the data is available only to the intended and authorized persons.
- **Integrity** – This goal means maintaining and assuring the accuracy and consistency of data. The function of integrity is to make sure that the data is reliable and is not changed by unauthorized persons.
- **Availability** – The function of availability in Network Security is to make sure that the data, network resources/services are continuously available to the legitimate users, whenever they require it.

Achieving Network Security

Ensuring network security may appear to be very simple. The goals to be achieved seem to be straightforward. But in reality, the mechanisms used to achieve these goals are highly complex, and understanding them involves sound reasoning.

International Telecommunication Union (ITU), in its recommendation on security architecture X.800, has defined certain mechanisms to bring the standardization in methods to achieve network security. Some of these mechanisms are –

- **En-cipherment** – This mechanism provides data confidentiality services by transforming data into not-readable forms for the unauthorized persons. This mechanism uses encryption-decryption algorithm with secret keys.
- **Digital signatures** – This mechanism is the electronic equivalent of ordinary signatures in electronic data. It provides authenticity of the data.
- **Access control** – This mechanism is used to provide access control services. These mechanisms may use the identification and authentication of an entity to determine and enforce the access rights of the entity.

Having developed and identified various security mechanisms for achieving network security, it is essential to decide where to apply them; both physically (at what location) and logically (at what layer of an architecture such as TCP/IP).

Security Mechanisms at Networking Layers

Several security mechanisms have been developed in such a way that they can be developed at a specific layer of the OSI network layer model.

- **Security at Application Layer** – Security measures used at this layer are application specific. Different types of application would need separate security measures. In order to ensure application layer security, the applications need to be modified.

It is considered that designing a cryptographically sound application protocol is very difficult and implementing it properly is even more challenging. Hence, application layer security mechanisms for protecting network communications are preferred to be only standards-based solutions that have been in use for some time.

An example of application layer security protocol is Secure Multipurpose Internet Mail Extensions (S/MIME), which is commonly used to encrypt e-mail messages. DNSSEC is another protocol at this layer used for secure exchange of DNS query messages.

- **Security at Transport Layer** – Security measures at this layer can be used to protect the data in a single communication session between two hosts. The most common use for transport layer security protocols is protecting the HTTP and FTP session traffic. The Transport Layer Security (TLS) and Secure Socket Layer (SSL) are the most common protocols used for this purpose.

- **Network Layer** – Security measures at this layer can be applied to all applications; thus, they are not application-specific. All network communications between two hosts or networks can be protected at this layer without modifying any application. In some environments, network layer security protocol such as Internet Protocol Security (IPsec) provides a much better solution than transport or application layer controls because of the difficulties in adding controls to individual applications. However, security protocols at this layer provide less communication flexibility that may be required by some applications.

Incidentally, a security mechanism designed to operate at a higher layer cannot provide protection for data at lower layers, because the lower layers perform functions of which the higher layers are not aware. Hence, it may be necessary to deploy multiple security mechanisms for enhancing the network security.

In the following chapters of the tutorial, we will discuss the security mechanisms employed at different layers of OSI networking architecture for achieving network security.

7.3. Cryptography

Human being from ages had two inherent needs – (a) to communicate and share information and (b) to communicate selectively. These two needs gave rise to the art of coding the messages in such a way that only the intended people could have access to the information. Unauthorized people could not extract any information, even if the scrambled messages fell in their hand.

The art and science of concealing the messages to introduce secrecy in information security is recognized as cryptography.

The word ‘cryptography’ was coined by combining two Greek words, ‘Krypto’ meaning hidden and ‘graphene’ meaning writing.

History of Cryptography

The art of cryptography is considered to be born along with the art of writing. As civilizations evolved, human beings got organized in tribes, groups, and kingdoms. This led to the emergence of ideas such as power, battles, supremacy, and politics. These ideas further fueled the natural need of people to communicate secretly with selective recipient which in turn

ensured the continuous evolution of cryptography as well.

The roots of cryptography are found in Roman and Egyptian civilizations.

Context of Cryptography

Cryptology, the study of cryptosystems, can be subdivided into two branches –

- Cryptography
- Cryptanalysis

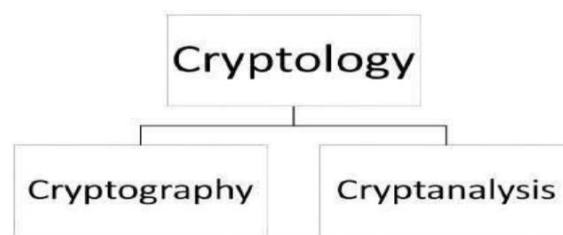


Figure 7. 1 Cryptology branches

What is Cryptography?

Cryptography is the art and science of making a cryptosystem that is capable of providing information security.

Cryptography deals with the actual securing of digital data. It refers to the design of mechanisms based on mathematical algorithms that provide fundamental information security services. You can think of cryptography as the establishment of a large toolkit containing different techniques in security applications.

What is Cryptanalysis?

The art and science of breaking the cipher text is known as cryptanalysis.

Cryptanalysis is the sister branch of cryptography and they both co-exist. The cryptographic process results in the cipher text for transmission or storage. It involves the study of cryptographic mechanism with the intention to break them. Cryptanalysis is also used during the design of the new cryptographic techniques to test their security strengths.

Security Services of Cryptography

The primary objective of using cryptography is to provide the following four fundamental information security services. Let us now see the possible goals intended to be fulfilled by cryptography.

Confidentiality

Confidentiality is the fundamental security service provided by cryptography. It is a security service that keeps the information from an unauthorized person. It is sometimes referred to as privacy or secrecy. Confidentiality can be achieved through numerous means starting from physical securing to the use of mathematical algorithms for data encryption.

Data Integrity

It is security service that deals with identifying any alteration to the data. The data may get modified by an unauthorized entity intentionally or accidentally. Integrity service confirms that whether data is intact or not since it was last created, transmitted, or stored by an authorized user.

Data integrity cannot prevent the alteration of data, but provides a means for detecting whether data has been manipulated in an unauthorized manner.

Authentication

Authentication provides the identification of the originator. It confirms to the receiver that the data received has been sent only by an identified and verified sender.

Authentication service has two variants –

- Message authentication identifies the originator of the message without any regard router or system that has sent the message.
- Entity authentication is assurance that data has been received from a specific entity, say a particular website.

Apart from the originator, authentication may also provide assurance about other parameters related to data such as the date and time of creation/transmission.

Non-repudiation

It is a security service that ensures that an entity cannot refuse the ownership of a previous commitment or an action. It is an assurance that the original creator of the data cannot deny the creation or transmission of the said data to a recipient or third party.

Non-repudiation is a property that is most desirable in situations where there are chances of a dispute over the exchange of data. For example, once an order is placed electronically, a purchaser cannot deny the purchase order, if non-repudiation service was enabled in this transaction.

Cryptography Primitives

Cryptography primitives are nothing but the tools and techniques in Cryptography that can be selectively used to provide a set of desired security services –

- Encryption
- Hash functions
- Message Authentication codes (MAC)
- Digital Signatures

The following table shows the primitives that can achieve a particular security service on their own.

Primitives Service ➡ ↓	Encryption	Hash Function	MAC	Digital Signature
Confidentiality	Yes	No	No	No
Integrity	No	Sometimes	Yes	Yes
Authentication	No	No	Yes	Yes
Non Reputation	No	No	Sometimes	Yes

Table 7.1: Security services of cryptography

A cryptosystem is an implementation of cryptographic techniques and their accompanying infrastructure to provide information security services. A cryptosystem is also referred to as a **cipher system**.

Let us discuss a simple model of a cryptosystem that provides confidentiality to the information being transmitted. This basic model is depicted in the illustration below –

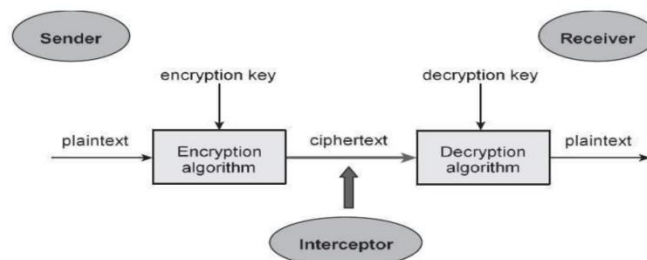


Figure 7. 2 Simple model of cryptography

The illustration shows a sender who wants to transfer some sensitive data to a receiver in such a way that any party intercepting or eavesdropping on the communication channel cannot extract the data.

The objective of this simple cryptosystem is that at the end of the process, only the sender and the receiver will know the plaintext.

Components of a Cryptosystem

The various components of a basic cryptosystem are as follows –

- **Plaintext.** It is the data to be protected during transmission.
- **Encryption Algorithm.** It is a mathematical process that produces a ciphertext for any given plaintext and encryption key. It is a cryptographic algorithm that takes plaintext and an encryption key as input and produces a ciphertext.
- **Ciphertext.** It is the scrambled version of the plaintext produced by the encryption algorithm using a specific the encryption key. The ciphertext is not guarded. It flows on public channel. It can be intercepted or compromised by anyone who has access to the communication channel.
- **Decryption Algorithm,** It is a mathematical process, that produces a unique plaintext for any given ciphertext and decryption key. It is a cryptographic algorithm that takes a ciphertext and a decryption key as input, and outputs a plaintext. The decryption algorithm essentially reverses the encryption algorithm and is thus closely related to it.
- **Encryption Key.** It is a value that is known to the sender. The sender inputs the encryption key into the encryption algorithm along with the plaintext in order to compute the ciphertext.
- **Decryption Key.** It is a value that is known to the receiver. The decryption key is related to the encryption key, but is not always identical to it. The receiver inputs the decryption key into the decryption algorithm along with the ciphertext in order to compute the plaintext.

For a given cryptosystem, a collection of all possible decryption keys is called a **key space**.

An **interceptor** (an attacker) is an unauthorized entity who attempts to determine the plaintext.

He can see the ciphertext and may know the decryption algorithm. He, however, must never know the decryption key.

7.4. Types of Cryptosystems

Fundamentally, there are two types of cryptosystems based on the manner in which encryption/decryption is carried out in the system –

- Symmetric Key Encryption
- Asymmetric Key Encryption

The main difference between these cryptosystems is the relationship between the encryption and the decryption key. Logically, in any cryptosystem, both the keys are closely associated. It is practically impossible to decrypt the ciphertext with the key that is unrelated to the encryption key.

Symmetric Key Encryption

The encryption process where same keys are used for encrypting and decrypting the information is known as Symmetric Key Encryption.

The study of symmetric cryptosystems is referred to as symmetric cryptography. Symmetric cryptosystems are also sometimes referred to as secret key cryptosystems.

A few well-known examples of symmetric key encryption methods are – Digital Encryption Standard (DES), Triple-DES (3DES), IDEA, and BLOWFISH.

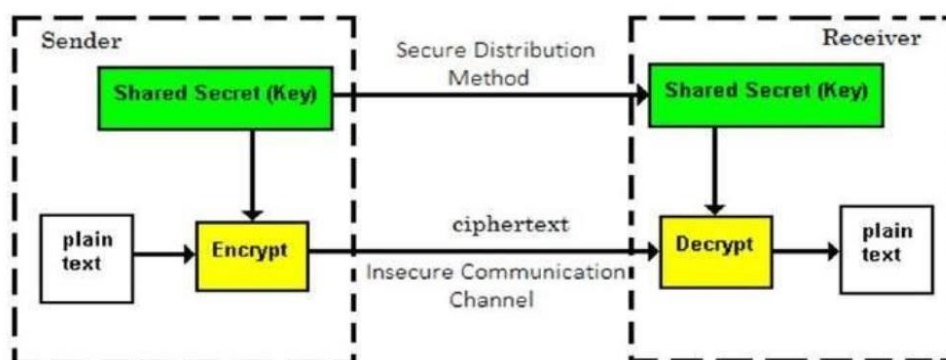


Figure 7.3 Symmetric key encryption

Prior to 1970, all cryptosystems employed symmetric key encryption. Even today, its relevance is very high and it is being used extensively in many cryptosystems. It is very unlikely that this encryption will fade away, as it has certain advantages over asymmetric key encryption.

The salient features of cryptosystem based on symmetric key encryption are –

- Persons using symmetric key encryption must share a common key prior to exchange of information.
- Keys are recommended to be changed regularly to prevent any attack on the system.
- A robust mechanism needs to exist to exchange the key between the communicating parties. As keys are required to be changed regularly, this mechanism becomes expensive and cumbersome.
- In a group of n people, to enable two-party communication between any two persons, the number of keys required for group is $n \times (n - 1)/2$.
- Length of Key (number of bits) in this encryption is smaller and hence, process of encryption-decryption is faster than asymmetric key encryption.
- Processing power of computer system required to run symmetric algorithm is less.

Challenge of Symmetric Key Cryptosystem

There are two restrictive challenges of employing symmetric key cryptography.

- Key establishment – before any communication, both the sender and the receiver need to agree on a secret symmetric key. It requires a secure key establishment mechanism in place.
- Trust Issue – since the sender and the receiver use the same symmetric key, there is an implicit requirement that

the sender and the receiver 'trust' each other. For example, it may happen that the receiver has lost the key to an attacker and the sender is not informed.

These two challenges are highly restraining for modern day communication. Today, people need to exchange information with non-familiar and non-trusted parties. For example, a communication between online seller and customer. These limitations of symmetric key encryption gave rise to asymmetric key encryption schemes.

Asymmetric Key Encryption

The encryption process where **different keys are used for encrypting and decrypting the information** is known as Asymmetric Key Encryption. Though the keys are different, they are mathematically related and hence, retrieving the plaintext by decrypting ciphertext is feasible. The process is depicted in the following illustration –

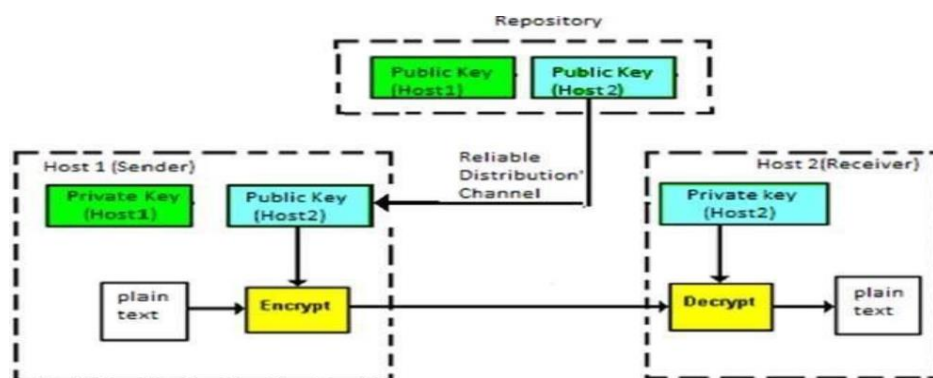


Figure 7. 4 Asymmetric key encryption

Asymmetric Key Encryption was invented in the 20th century to come over the necessity of preshared secret key between communicating persons. The salient features of this encryption scheme are as follows –

- Every user in this system needs to have a pair of dissimilar keys, **private key and public key**. These keys are mathematically related – when one key is used for encryption, the other can decrypt the ciphertext back to the original plaintext.
- It requires to put the public key in public repository and the private key as a wellguarded secret. Hence, this scheme of encryption is also called **Public Key Encryption**.
- Though public and private keys of the user are related, it is computationally not feasible to find one from another. This is a strength of this scheme.
- When Host1 needs to send data to Host2, he obtains the public key of Host2 from repository, encrypts the data, and transmits.
- Host2 uses his private key to extract the plaintext.
- Length of Keys (number of bits) in this encryption is large and hence, the process of encryption-decryption is slower than symmetric key encryption.
- Processing power of computer system required to run asymmetric algorithm is higher.

Symmetric cryptosystems are a natural concept. In contrast, public-key cryptosystems are quite difficult to comprehend.

Challenge of Public Key Cryptosystem

Public-key cryptosystems have one significant challenge – the user needs to trust that the public key that he is using in communications with a person really is the public key of that person and has not been spoofed by a malicious third party. This is usually accomplished through a Public Key Infrastructure (PKI) consisting a trusted third party. The third party securely manages and attests to the authenticity of public keys.

When the third party is requested to provide the public key for any communicating person X, they are trusted to provide the correct public key.

The third party satisfies itself about user identity by the process of attestation, notarization, or some other process – that X is the one and only, or globally unique, X. The most common method of making the verified public keys available is to embed them in a certificate which is digitally signed by the trusted third party.

Relation between Encryption Schemes

	Symmetric Cryptosystems	Public Key Cryptosystems
Relation between Keys	Same	Different, but mathematically related
Encryption Key	Symmetric	Public
Decryption Key	Symmetric	Private

Table 7.2: Summary of basic key cryptosystems

Due to the advantages and disadvantage of both the systems, symmetric key and public-key cryptosystems are often used together in the practical information security systems.

Private Key

In Private key, the same key (secret key) is used for encryption and decryption. In this key is symmetric because the only key is copy or share by another party to decrypt the cipher text. It is faster than the public key cryptography.

Public Key

In Public key, two keys are used one key is used for encryption and another key is used for decryption. One key (public key) is used for encrypt the plain text to convert it into cipher text and another key (private key) is used by receiver to decrypt the cipher text to read the message.

Now, we see the difference between them:

S.NO	Private Key	Public Key
1.	Private key is faster than public key.	It is slower than private key.
2.	In this, the same key (secret key) and algorithm is used to encrypt	In public key cryptography, two keys are used, one key is used for encryption
	and decrypt the message.	and while the other is used for decryption.
3.	In private key cryptography, the key is kept as a secret.	In public key cryptography, one of the two keys is kept as a secret.
4.	Private key is Symmetrical because there is only one key that is called secret key.	Public key is Asymmetrical because there are two types of key: private and public key.
5.	In this cryptography, sender and receiver need to share the same key.	In this cryptography, sender and receiver does not need to share the same key.
6.	In this cryptography, the key is private.	In this cryptography, public key can be public and private key is private.

Table 7.3: difference between public key and private key

7.5. Firewalls

A firewall is a **network security** device that monitors incoming and outgoing network traffic and permits or blocks data **packets** based on a set of security rules. Its purpose is to establish a barrier between your internal network and incoming traffic from external sources (such as the internet) in order to block malicious traffic like viruses and hackers.

Firewalls carefully analyze incoming traffic based on pre-established rules and filter traffic coming from unsecured or suspicious sources to prevent attacks. Firewalls guard traffic at a computer's entry point, called ports, which is where information is exchanged with external devices. For example, "Source address 172.18.1.1 is allowed to reach destination 172.18.2.1 over port 22."

Think of IP addresses as houses, and port numbers as rooms within the house. Only trusted people (source addresses) are allowed to enter the house (destination address) at all—then it's further filtered so that people within the house are only allowed to access certain rooms (destination ports), depending on if they're the owner, a child, or a guest. The owner is allowed to any room (any port), while children and guests are allowed into a certain set of rooms (specific ports)

7.6. Virtual Private Network

VPN stands for "**Virtual Private Network**" and describes the opportunity to establish a protected network connection when using public networks. VPNs encrypt your internet traffic and disguise your online identity. This makes it more difficult for third parties to track your activities online and steal data. The encryption takes place in **real time**.

Activity 7.2

- What do you know about Virtual private network?
- Discuss with your classmate how VPN works?
- List some very popular VPN servers?

How does a VPN work?

A VPN hides your IP address by letting the network redirect it through a specially configured remote server run by a VPN host. This means that if you surf online with a VPN, the VPN server becomes the source of your data. This means your Internet Service Provider (ISP) and other third parties cannot see which websites you visit or what data you send and receive online. A VPN works like a filter that turns all your data into "gibberish". Even if someone were to get their hands on your data, it would be useless.

What are the benefits of a VPN connection?

A VPN connection disguises your data traffic online and protects it from external access. Unencrypted data can be viewed by anyone who has network access and wants to see it. With a VPN, hackers and cyber criminals can't decipher this data.

Secure encryption: To read the data, you need an encryption key. Without one, it would take millions of years for a computer to decipher the code in the event of a brute force attack. With the help of a VPN, your online activities are hidden even on public networks.

Disguising your whereabouts : VPN servers essentially act as your proxies on the internet.

Because the demographic location data comes from a server in another country, your actual location cannot be determined. In addition, most VPN services do not store logs of your activities. Some providers, on the other hand, record your behavior, but do not pass this information on to third parties. This means that any potential record of your user behavior remains permanently hidden.

Access to regional content: Regional web content is not always accessible from everywhere.

Services and websites often contain content that can only be accessed from certain parts of the world. Standard connections use local servers in the country to determine your location. This means that you cannot access content at home while traveling, and you cannot access international content from home. With **VPN location spoofing**, you can switch to a server to another country and effectively “change” your location.

Secure data transfer: If you work remotely, you may need to access important files on your company’s network. For security reasons, this kind of information requires a secure connection.

To gain access to the network, a VPN connection is often required. VPN services connect to private servers and use encryption methods to reduce the risk of data leakage.

7.7. Chapter Seven Review Questions

- What are network security issues?
- Discuss on three pillars of network security?
- What is cryptography?
- What is Digital signature means?
- What are Symmetric key and asymmetric key encryption techniques?
- What is Public key and private key means?

